

Dendritic growth and synaptic organization from activity-independent cues and local activity-dependent plasticity

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eLife Assessment

This **important** work investigates how two distinct processes, morphological changes and synaptic plasticity, contribute to the final shape of neuronal dendrites and the spatial structure of their synaptic inputs. The modelling is **convincing** and could be broadly applied to other similar questions. The work will be of interest to neuroscientists studying dendritic development and connectivity at a single-cell level.

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Abstract

Dendritic branching and synaptic organization shape single-neuron and network computations. How they emerge simultaneously during brain development as neurons become integrated into functional networks is still not mechanistically understood. Here, we propose a mechanistic model in which dendrite growth and the organization of synapses arise from the interaction of activity-independent cues from potential synaptic partners and local activity-dependent synaptic plasticity. Consistent with experiments, three phases of dendritic growth – overshoot, pruning, and stabilization – emerge naturally in the model. The model generates stellate-like dendritic morphologies that capture several morphological features of biological neurons under normal and perturbed learning rules, reflecting biological variability. Model-generated dendrites have approximately optimal wiring length consistent with experimental measurements. In addition to establishing dendritic morphologies, activity-dependent plasticity rules organize synapses into spatial clusters according to the correlated activity they experience. We demonstrate that a trade-off between activity-dependent and -independent factors influences dendritic growth and synaptic location throughout development, suggesting that early developmental variability can affect mature morphology and synaptic function. Therefore, a single mechanistic model can capture dendritic growth and account for the synaptic organization of correlated inputs during development. Our work suggests concrete mechanistic components underlying the emergence of dendritic morphologies and synaptic formation and removal in function and

dysfunction, and provides experimentally testable predictions for the role of individual components.

Introduction

The dendrites of a neuron are intricately branched structures that receive electrochemical stimulation from other neurons. The morphology of dendrites determines the location of synaptic contacts with other neurons, thereby constraining single-neuron computations. During development, the dendrites of many neurons grow simultaneously and become integrated into neural circuits. The development of dendrites is highly dynamic; iterated addition and retraction of branches allow these dendrites to probe various potential synaptic partners before stabilizing (Cline, 2016 [↗](#); Richards et al., 2020 [↗](#)). Many intrinsic and extrinsic factors underlie the dynamics of dendritic development. In any given neuron, the intrinsic expression of specific genes controls many morphological aspects, including the orientation of the dendrite in the cortex, the general abundance of dendritic branching, and the timing of growth onset (Puram and Bonni, 2013 [↗](#)). In contrast, extrinsic signaling exerts precise control over the detailed dynamics of dendrite development through various mechanisms, including activity-dependent cell-to-cell interactions and molecular signaling (Polleux et al., 2016 [↗](#)).

Although many signaling molecules affect dendrite development, brain-derived neurotrophic factor (BDNF) and its immature predecessor proBDNF are particularly crucial in the central nervous system (Lu et al., 2005 [↗](#)). While exposure to BDNF leads to larger dendrites with a higher density of synapses (McAllister et al., 1995 [↗](#); Tyler and Pozzo-Miller, 2001 [↗](#)), exposure to proBDNF leads to smaller dendrites with fewer synapses (Koshimizu et al., 2009 [↗](#); Yang et al., 2014 [↗](#)). Furthermore, the precise balance of BDNF and proBDNF is essential for the organization of synapses into clusters during development (Kirchner and Gjorgjieva, 2021 [↗](#); Winnubst et al., 2015 [↗](#); Kleindienst et al., 2011 [↗](#); Niculescu et al., 2018 [↗](#)). Interestingly, synaptic activity triggers the cleaving of proBDNF into BDNF (Je et al., 2012 [↗](#)), providing a mechanistic link between the molecular factors driving dendrite maturation and neural activity.

Activity-dependent factors are equally important for driving dendritic growth. As the sensory periphery is immature during early postnatal development, when many dendrites grow (Leighton and Lohmann, 2016 [↗](#)), many developing circuits generate their own spontaneous activity. The rich spatiotemporal structure of spontaneous activity instructs the formation, removal and change in the strength of synaptic inputs (Sretavan et al., 1988 [↗](#); Sakai, 2020 [↗](#)) and triggers the stabilization or retraction of entire dendritic branches (Riccomagno and Kolodkin, 2015 [↗](#); Lohmann et al., 2002 [↗](#)). Although blocking spontaneous activity does not result in a grossly different dendrite morphology, the density and specificity of synaptic connections are strongly perturbed (Campbell et al., 1997 [↗](#); Ultanir et al., 2007 [↗](#)), highlighting the instructive effect of spontaneous activity on dendritic development (Crair, 1999 [↗](#)).

An influential hypothesis tying the extrinsic signaling factors underlying dendritic development is the synaptotrophic hypothesis (Vaughn, 1989 [↗](#)). According to this hypothesis, a growing dendrite preferentially extends to regions where it is likely to find synaptic partners. Once a dendrite finds such a partner, a synaptic contact forms, anchors the developing dendrite, and serves as an outpost for further dendritic growth. In contrast, loss of synaptic input to the dendrite can lead to retraction unless the remaining synapses stabilize the branch (Lohmann et al., 2002 [↗](#); Niell et al., 2004 [↗](#); Haas et al., 2006 [↗](#); Cline and Haas, 2008 [↗](#); Riccomagno and Kolodkin, 2015 [↗](#); Cline, 2016 [↗](#)). However, elaborate dendrites with morphologically defined synapses can also emerge without any synaptic transmission (Verhage et al., 2000 [↗](#); Cijssouw et al., 2014 [↗](#); Ferreira Castro et al., 2020 [↗](#), 2023 [↗](#)), suggesting that synaptic activity influences dendritic growth, but it is not the

only driving force. Despite significant interest in the synaptotrophic hypothesis, we still lack a mechanistic understanding of how activity-dependent and -independent factors combine to shape development.

To investigate interactions between known signaling factors and synthesize information from different experimental results, computational models of dendrite development provide a fruitful direction to explore how different mechanisms can generate realistic dendritic morphologies (Cuntz, 2016 [↗](#); Ferreira Castro et al., 2020 [↗](#)). Previous approaches include modeling dendritic development with random branching (Kliemann, 1987 [↗](#)) or as a reaction-diffusion system (Luczak, 2006 [↗](#)), implementing activity-independent growth cones that detect molecular gradients (van Veen and van Pelt, 1992 [↗](#); Torben-Nielsen and De Schutter, 2014 [↗](#)), or building dendrites as the solution to an optimal wiring problem (Cuntz et al., 2010 [↗](#)). Although these approaches are capable of generating dendritic structures that closely replicate the statistical properties of both developing and mature biological dendrites (Koene et al., 2009 [↗](#); Cuntz, 2016 [↗](#)), they provide limited insight into the interaction between dendritic growth, synaptogenesis, and the local activity-dependent organization of synaptic inputs. Consequently, the association between morphological variability and electro-physiological (Gouwens et al., 2020 [↗](#); Scala et al., 2021 [↗](#)) or functional (Poirazi and Mel, 2001 [↗](#); Poirazi et al., 2003 [↗](#); Park et al., 2019 [↗](#); Poirazi and Papoutsis, 2020 [↗](#)) characteristics of synaptic and dendritic attributes remains obscure.

Here, we propose a mechanistic computational model for cortical dendritic development for dendrite growth and synapse formation, stabilization, and elimination based on reciprocal interactions between activity-independent growth signals and spontaneous activity. Starting from neuronal somata distributed in a flat sheet of the cortex, spatially distributed potential synapses drive the growth of stellate-like dendrites through elongation and branching by activity-independent cues. Upon contact, synaptic connections form and stabilize or disappear according to a local activity-dependent learning rule inspired by neurotrophin interactions based on correlated patterns of spontaneous activity (Kirchner and Gjorgjieva, 2021 [↗](#)). Consistent with the synaptotrophic hypothesis, the stability of a dendritic branch depends on the stability of its synaptic contacts, and the branch likely retracts after substantial synaptic pruning. The resulting dynamic system naturally leads to the emergence of three distinct phases of dendrite development: 1) an initial over-shoot phase characterized by dendrite growth and synapse formation, 2) a pruning phase during which the learning rule prunes poorly synchronized synapses, and 3) a stabilization phase during which morphologically stable dendrites emerge from the balancing of growth and retraction. Varying model parameters in biologically realistic ranges produce changes in dendrite length and synapse density consistent with experiments. Our mechanistic model generates dendrites with approximately optimal wiring length, which is a widely used criterion for evaluating dendritic morphology (Cuntz et al., 2010 [↗](#), 2012 [↗](#); Chklovskii et al., 2002 [↗](#)). At the same time, the model leads to the activity-dependent emergence of functional synaptic organization and input selectivity. Therefore, our mechanistic modeling framework for the growth and stabilization of dendritic morphologies and the simultaneous synaptic organization according is ideally suited for making experimental predictions about the effect of perturbing specific model components on the resulting dendritic morphologies and synaptic placement.

Results

We built a computational model of activity-dependent dendrite growth during development based on synapse formation, stabilization, and elimination. We focused on the basal stellate-like dendrites of cortical pyramidal neurons, which primarily extend laterally within a layer of the cortex (Larkman and Mason, 1990 [↗](#)) and receive numerous feedforward and recurrent inputs (Rossi et al., 2019 [↗](#); Iacaruso et al., 2017 [↗](#)). Stellate morphologies are found in many types of neurons, especially in the somatosensory cortex, including interneurons and layer 4 spiny stellate

cells, which are the main recipients of thalamic input and play a key role in sensory processing (Schubert et al., 2003 [↗](#); Marques-Smith et al., 2016 [↗](#); Scala et al., 2019 [↗](#)). To investigate the impact of synapse formation on dendrite development, we modeled several neuronal somata and potential synapses in a flat sheet of cortex (**Figure 1a** [↗](#)). Potential synapses represent locations in the cortex where an axon can form a synapse with a nearby dendrite (Stepanyants and Chklovskii, 2005 [↗](#)). The model consists of two components: An activity-independent component that directly governs branch growth and retraction; and an activity-dependent component that governs synaptic organization and thus indirectly affects branch stability. Inspired by the synaptotrophic hypothesis (Vaughn, 1989 [↗](#)), we mimicked the effect of activity-independent molecular signaling by letting each potential synapse release diffusive signaling molecules that attract the growing dendrite (**Figure 1b** [↗](#), **Figure 1–Figure Supplement 1** [↗](#), **Figure 1–Figure Supplement 2** [↗](#)). In addition, during development and before the on-set of sensory experience, neural circuits generate patterned spontaneous activity (Blankenship and Feller, 2010 [↗](#); Ackman and Crair, 2014 [↗](#)). Therefore, to model the structured spontaneous activity exhibited by different axons (Scholl et al., 2017 [↗](#); Iacaruso et al., 2017 [↗](#)), we randomly divided the potential synapses into different activity groups that receive inputs correlated within a group but uncorrelated between the groups (see Methods). Each group represents synapses from the same presynaptic neuron or neurons that experience correlated presynaptic activity.

Attracted by synapses which release growth factors and independent of neural activity, dendrites in our model extend from the soma towards the closest potential synapse, where they establish a synapse and temporarily stabilize. (**Figure 1b** [↗](#), **Figure 1–Figure Supplement 2** [↗](#)). We assumed that the dendrites could not overlap based on experimental data (Grueber and Sagasti, 2010 [↗](#)); therefore, the dendrites in the model retract, for example, when further growth would require self-overlap. Once a synapse is formed, we modeled that its strength changes according to a local activity-dependent plasticity rule (Kirchner and Gjorgjieva, 2021 [↗](#)) (**Figure 1c** [↗](#)). The learning rule induces synaptic potentiation when presynaptic and local postsynaptic activity coincide. Conversely, it induces synaptic depression when local postsynaptic activity occurs in the dendrite independently of presynaptic stimulation, usually due to the activation of a neighboring synapse (see Methods and the 'offset' constant below),

$$\Delta \text{weight} = \text{post} \times (\text{pre} - \text{offset}). \quad (1)$$

As shown previously, this rule generates synaptic distance-dependent competition, where nearby synapses affect each other more than distant synapses, and correlation-dependent cooperation, where neighboring synchronized synapses stabilize. In contrast, neighboring desynchronized synapses depress (Kirchner and Gjorgjieva, 2021 [↗](#)). In our model, we assumed that when a synapse depresses below a threshold, it decouples from the dendrite, and the corresponding branch retracts successively to the nearest stable synapse, branch point, or the soma (**Figure 1b** [↗](#), **Figure 1–Figure Supplement 2** [↗](#)). After removal, the vacated synapse turns into a potential synapse again, attracting other growing branches. Thus, a developing dendrite in our model acquires its arborization through the attraction to signaling molecules released by potential synapses and the repeated activity-dependent formation, stabilization, and removal of synapses.

Dendrite development through balancing growth and retraction

After specifying the rules governing the growth of individual dendritic branches, we investigated dendritic development on longer timescales. When growing dendrites according to our proposed growth rule based on the attraction of signaling molecules and spontaneous activity-dependent synaptic refinements (**Figure 1** [↗](#)), we found that dendrites form several stems, i.e. branches which start directly at the soma, and rapidly expand outwards (**Figure 2a** [↗](#)). After an initial phase of rapid expansion, we observed that growth rapidly attenuates and dendritic length stabilizes (**Figure 2b** [↗](#)). This stability is achieved when the expansion and retraction of the dendrite are balanced (**Figure 2c**). To investigate whether the stability in total length also corresponds to stability in dendritic morphology, we quantified morphological stability as the

Figure 1.

A model of dendritic growth for a cortical pyramidal neuron driven by activity-independent and -dependent mechanisms.

(a) Schematic of the soma of a pyramidal neuron (orange triangle) with 12 randomly distributed potential synapses from presynaptic axons (circles) with correlated activity patterns indicated by color. (b) Schematic of activity-independent and -dependent mechanisms. Soma and synapses correspond to box in a. Signaling molecules diffusing from potential synapses (1) attract dendrite growth and promote synapse formation (2) independent of firing pattern (3). Over time, poorly synchronized synapses depress and are pruned from the dendrite (4), while well-synchronized synapses remain stable (5). After a branch retracts, the dendrite is less sensitive to the growth field at that location (5). (c) Change in weight of one synapse (green) following the stimulation of itself (green bolt) and of two nearby synapses (purple bolts). Left: Schematic of the developing dendrite from b with bolts indicating synaptic activation. Right: Presynaptic accumulator (top), postsynaptic accumulator (middle), and change in synaptic weight (bottom) as a function of time (see Methods [Kirchner and Gjorgjieva \(2021\)](#) [for details of the plasticity rule](#)). Dashed line (bottom) indicates zero change.

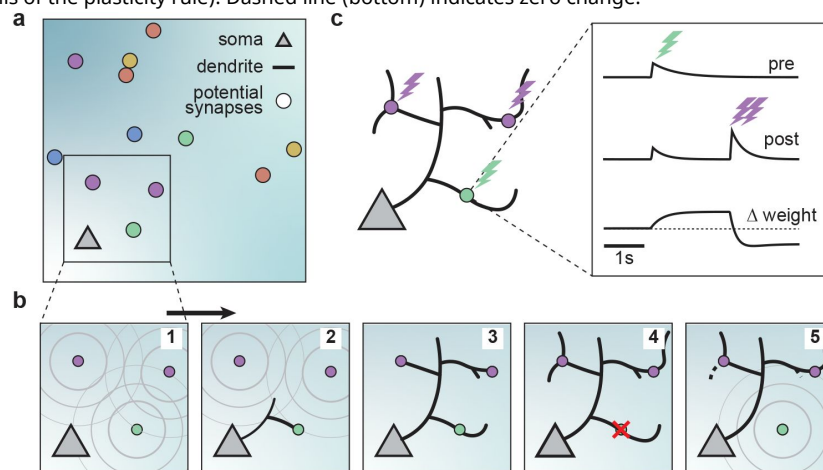
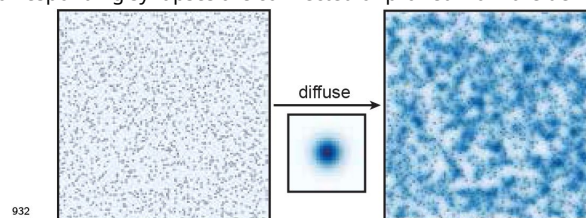


Figure 1–Figure supplement 1.

The growth field is similar to two-dimensional heat diffusion.

By iteratively convolving the potential synapses in the growth field with a Gaussian filter, the growth field reaches a steady-state that resembles a two-dimensional heat diffusion with point sources. Over time, individual point sources disappear and reappear to mimic when the corresponding synapses are connected or pruned from the dendrite.



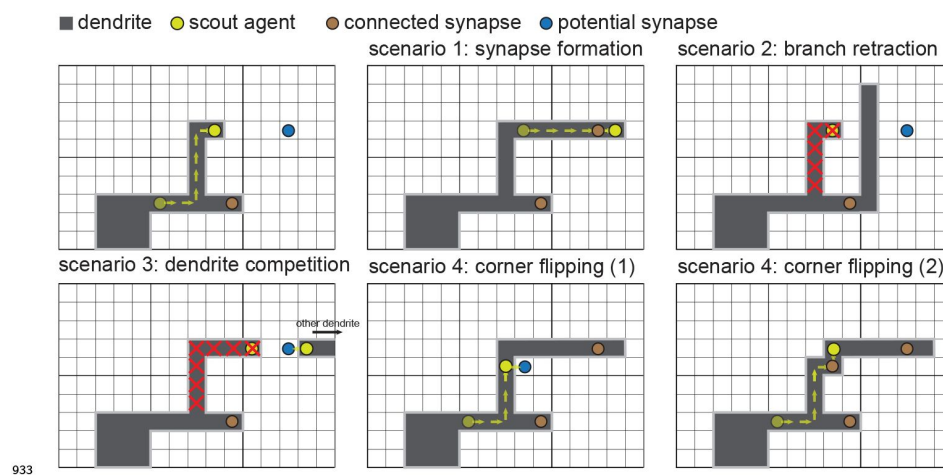


Figure 1-Figure supplement 2.

Four scenarios of asynchronous dendritic growth can be modeled as a scout agent representing the tip of a dendrite exploring a two-dimensional grid. A scout agent (yellow dot) has reached the location of a potential synapse (blue dot). Scenario 1: The scout agent will extend the dendrite and form a new synapse if nothing else happens. Scenario 2: To prevent overlap, if a second branch from the same dendrite blocks the path to the potential synapse, the original branch retracts. Scenario 3: If a branch from another dendrite reaches the potential synapse first, the original branch retracts. Scenario 4: If a new potential synapse becomes available adjacent to the dendrite (1) so that growth is not possible (since the branch cannot form immediately adjacent to two other parts of the dendrite), the corner flips (2), and the synapse forms.

pixel-wise correlation of a dendrite with itself 4.5 hours earlier, which is several orders of magnitude larger than the speed at which dendrites grow and retract in our model (see **Table 1** [↗](#)). Despite the residual amount of expansion and retraction, we showed that morphological stability increases rapidly and the dendritic morphology is already stable after the initial expansion phase (**Figure 2d** [↗](#)). Interestingly, such rapid stabilization of morphology has also been observed in the mouse visual cortex (Richards et al., 2020 [↗](#)) and the *Drosophila* larvae (Ferreira Castro et al., 2020 [↗](#)). Next, we quantified the Sholl diagram, the number of dendritic branches at a given distance from the soma, commonly used as a measure of dendritic complexity (Sholl, 1953 [↗](#); Binley et al., 2014 [↗](#); Bird and Cuntz, 2019 [↗](#)). The Sholl diagram of stabilized dendrites generated by our model is unimodal and qualitatively matches the Sholl diagram of developing basal dendrites from the mouse medial prefrontal cortex (**Figure 2e** [↗](#); data extracted from ref. Kroon et al., 2019 [↗](#), postnatal days 6-8), as well as the hippocampus (Kleindienst et al., 2011 [↗](#)). In summary, by combining activity-independent and -dependent dendritic growth mechanisms, our model produces dendrites that rapidly expand and stabilize by balancing growth and retraction.

Delayed activity-dependent plasticity produces a rapid increase in synapse density followed by pruning

Since our model couples dendritic growth with the formation and removal of synapses (**Figure 3a** [↗](#)), we next investigated how the number of connected synapses, which are necessary for the dendrite's stabilization, changes over time. As a result of the dendrite's rapid growth, we observed a rapid increase in the number of connected synapses (**Figure 3b,c** [↗](#)). In contrast to dendritic elongation, our findings indicate that the initial rapid increase in the number of connected synapses is subsequently followed by a transient phase of net synapse reduction before reaching a balance of synaptic additions and removals. (**Figure 3c** [↗](#)). This removal of established synapses resembles the postnatal removal of synapses observed in the mouse neocortex (Holtmaat et al., 2005 [↗](#)). To understand how the initial overshoot and subsequent removal of synapses emerge in our model, we computed the average synaptic weight of all synapses that eventually stabilize or are pruned (**Figure 3d** [↗](#)). We found that the delayed onset of synapse removal (**Figure 3c** [↗](#)) is due to the slow time scale of synaptic weight changes compared to the faster time scale of dendrite growth. Thus, the initial overshoot and subsequent removal of synapses observed in our model (**Figure 3b** [↗](#)) is due to the rapid formation relative to the delayed activity-dependent elimination of synapses. After the initial overshoot and pruning, dendritic branches in the model stay stable, with mainly small subbranches continuing to be refined (**Figure 3** [↗](#)-**Figure Supplement 1** [↗](#)). This stability in the model is achieved despite the number of potential synaptic partners remaining high (**Figure 3e** [↗](#)), indicating a balance between activity-independent and activity-dependent mechanisms. The dendritic growth and synaptic refinement dynamics are independent of the postsynaptic somata densities used in our simulations (**Figure 3**-**Figure Supplement 2** [↗](#)). Only the final arbor size and the number of connected synapses decrease with an increase in the density of the somata, while the timing of synaptic growth, overshoot and pruning remains the same (**Figure 3**-**Figure Supplement 2** [↗](#)).

The development of axons is concurrent with dendritic growth and highly dynamic Matsumoto et al. (2024) [↗](#). To address the impact of simultaneously growing axons, we implemented a simple form of axonal dynamics by allowing changes in the lifetime and location of potential synapses, originating from the axons of presynaptic partners (**Figure 3**-**Figure Supplement 3**). When potential synapses can move rapidly (median lifetime of 1.8 hours), the model dynamics are perturbed quite substantially, making it difficult for the dendrites to stabilize completely (**Figure 3**-**Figure Supplement 3c** [↗](#)). However, slowly moving potential synapses (median lifetime of 18 hours) still yield comparable results (**Figure 3**-**Figure Supplement 3** [↗](#)). The distance of movement significantly influenced results only when potential synaptic lifetimes were short. For extended lifetimes, the moving distance had a minor impact on the dynamics, predominantly affecting the time required for dendrites to stabilize. This was the result of synapses being able to transfer from

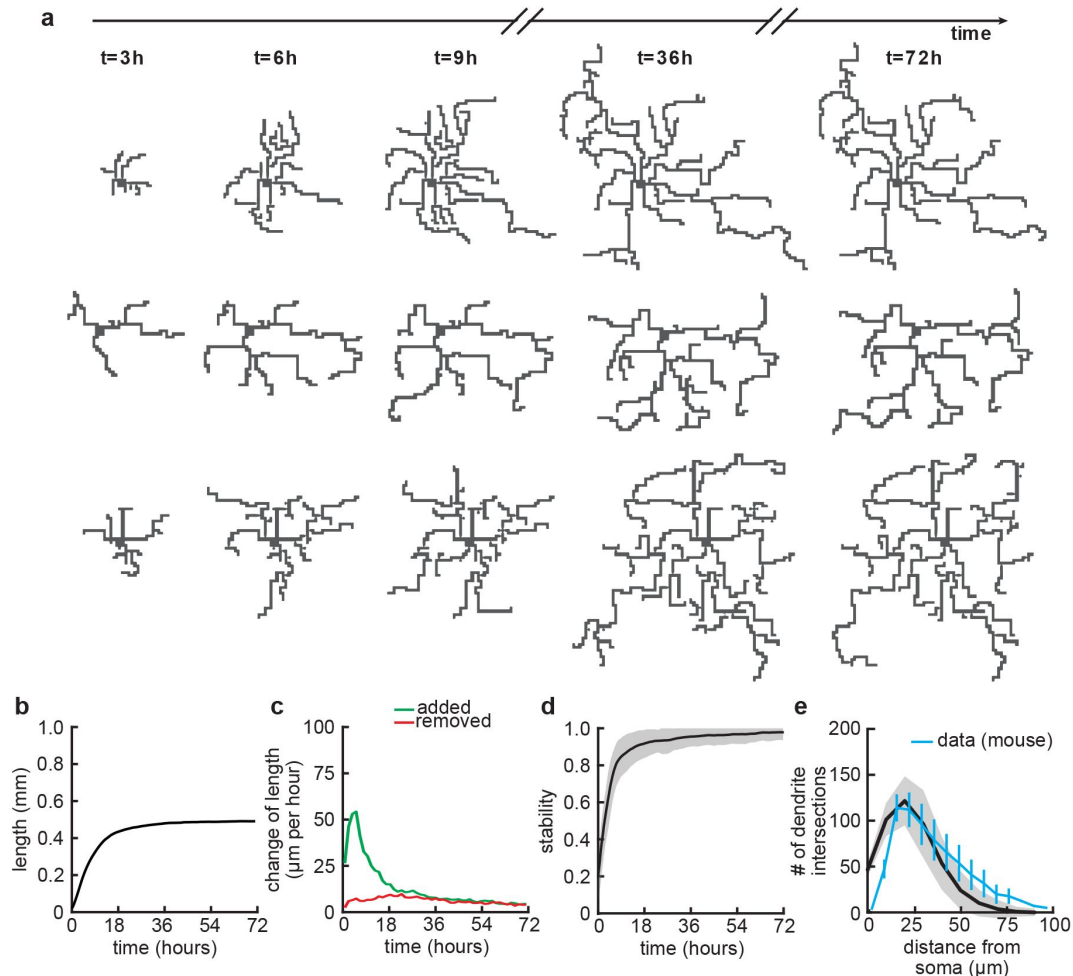


Figure 2.

Balanced growth and retraction generate morphologically stable dendrites.

a) Three example dendrites at five time points from our simulations. For clarity of presentation, connected synapses are not displayed. (b) Total length of dendritic tree as a function of time. (c) Length of dendrite added (green) and removed (red) as a function of time. (d) Morphological stability (correlation between the dendrite shape at time t and $t - 4.5$ hours) as a function of time. (e) Average number of dendrite intersections as a function of distance from the soma (the Sholl diagram). Data from basal dendrites in the developing mouse medial prefrontal cortex superimposed, normalized to the maximum (blue; ref. (Kroon et al., 2019)). All lines represent averages across 32 simulations with nine dendrites each. Shaded area indicates two standard deviations.

Table 1.

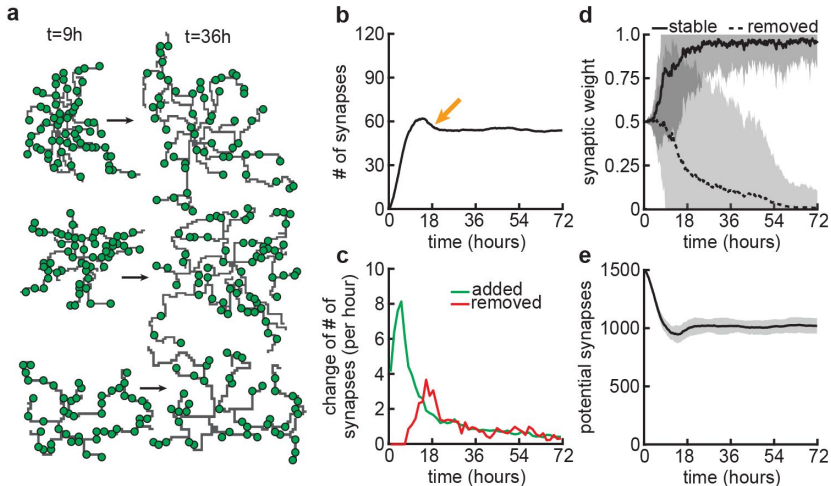
Parameters of the minimal plasticity model (Kirchner and Gjorgjieva, 2021) and the synaptotrophic growth model.

Parameter	Variable	Value
Synaptic efficacy time constant	τ_w	3000 time steps
Postsynaptic accumulator time constant	τ_u	300 time steps
Presynaptic accumulator time constant	τ_v	600 time steps
Constitutive percent of BDNF of total neurotrophins	η	46.5%
MMP9 efficiency constant	ϕ	$\frac{3}{50}$ per time step
Heterosynaptic offset	ρ	$\rho = \frac{2\eta-1}{2(1-\eta)}$
Minimal model synaptic efficacy time constant	τ_w	$\tau_w = \tau_w \frac{1}{2(1-\eta)}$
Standard deviation of calcium spread	σ_c	200 μm
Turnover threshold below which a synapse is replaced	W_{thr}	0.02
Firing rate of synapses	r_{total}	0.116 min^{-1}
Scout intervals and speed	$t_{\text{scout}} \cdot v_{\text{scout}}$	10 min, $0.18 \mu\text{m min}^{-1}$

Figure 3.

Synapse formation and removal predominate in distinct phases of dendrite development.

(a) Three examples of dendrites at the beginning ($t = 9$ hours) and end ($t = 72$ hours) of the simulation. Green circles indicate formed synapses. (b) Total number of connected synapses as a function of time. Orange arrow highlights overshoot and subsequent pruning. (c) Added (green) and pruned synapses (red) as a function of time. (d) Average synaptic weights of synapses that ultimately stabilize (solid black; final weight more than 0.5) or are removed (dashed black; final weight less than 0.5) as a function of time. (e) Number of available potential synapses as a function of time. All lines represent averages across 32 simulations with nine dendrites each. Shaded area indicates two standard deviations.



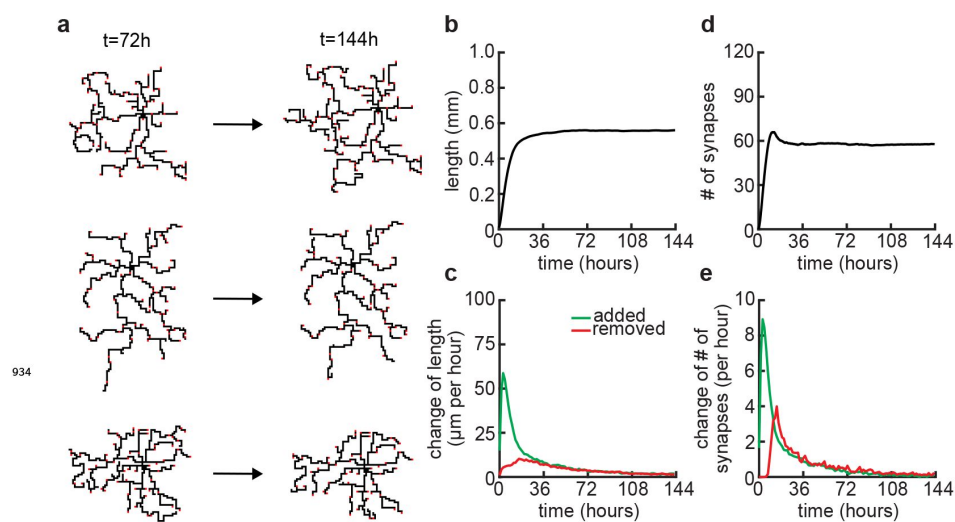


Figure 3-Figure supplement 1.

Dendritic arbors stay stable over longer simulations.

(a) Three examples of neurons at the end of the usual simulation time ($t = 72$ hours) and at double the usual simulation time ($t = 144$ hours). (b) Total length of dendritic tree as a function of time. (c) Length of dendrite added (green) and removed (red) as a function of time. (d) Total number of connected synapses as a function of time. (e) Added (green) and pruned synapses (red) as a function of time.

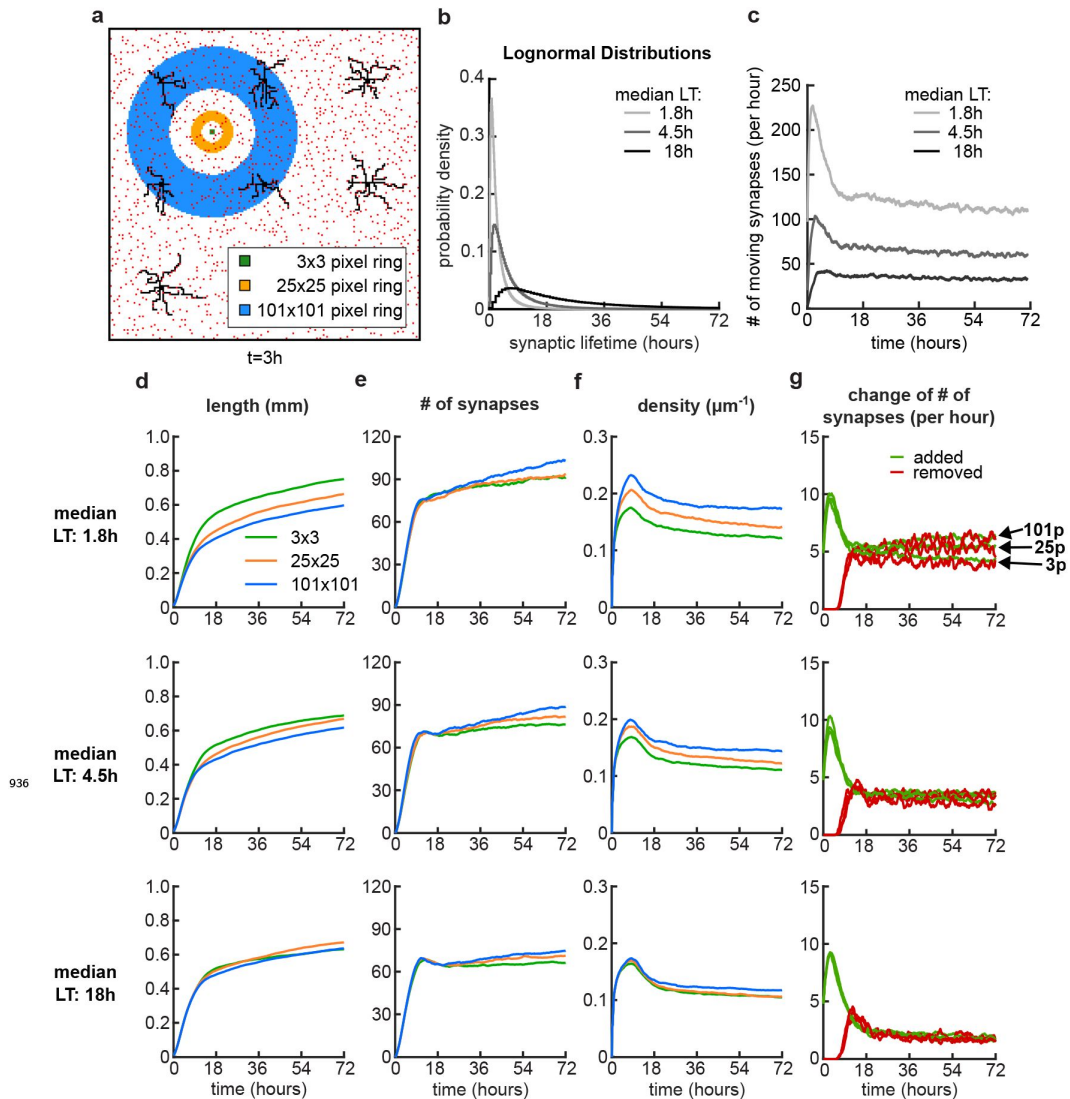


Figure 3-Figure supplement 2.

Somata density does not affect the timing of dendritic growth and pruning.

(a) Four examples of simulations with different numbers of somatas (from left to right: 9, 13, 16 and 25 somata) in the same simulated area at the beginning ($t = 3$ hours) and end ($t = 72$ hours) of simulation time. (b) Total length of dendritic tree as a function of time. (c) Total number of connected synapses as a function of time. (d) Total density of synapses on the dendrite as a function of time.

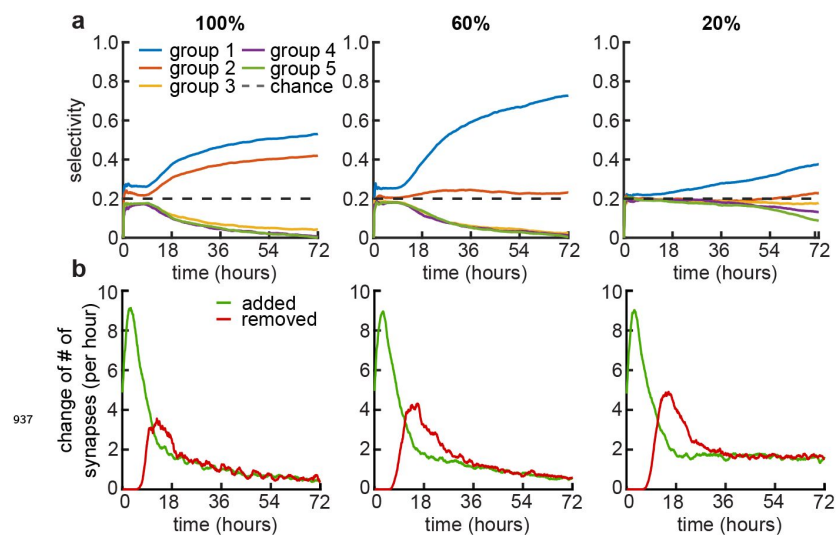


Figure 3-Figure supplement 3.

Dynamic potential synapses with log-normally distributed lifetimes.

(a) Nine dendrites (black) and potential synapses (red) after 3 hours of simulation time with a schematic of the pixel distance rings for the movement of an example potential synapse in the background. The synapse is located at the center of the rings and can move to any of the pixel locations of the specified ring, where there is no other synapse or dendrite located. (b) The synaptic lifetimes (LT) for each potential synapse are sampled from lognormal distributions with three different median LT (variance is always 1). (c) Number of moving synapses per hour over time for the three different distributions. (d-g) Dendrite length (d), number of synapses (e), density of connected synapses (f) and change of number of synapses (g) over time, for the median LTs 1.8 hours (top), 4.5 hours (center) and 18 hours (bottom). (g) At the top plot, the arrows highlight where the traces for the added (green) and removed (red) synapses end for different ring distances.

one dendrite to another, potentially forming new long-lasting connections even at advanced stages of synaptic refinement. In sum, our results show that potential axonal dynamics only affect the stability of our model when these dynamics are much faster than dendritic growth.

Activity-dependent competition between synapses produces input selectivity and synaptic organization

Next, we asked whether the stabilization of dendrites might be supported by the emergence of organization of connected synapses. First, we compared the synapses connected to dendrites at the apex of the overshoot phase (peak in [Figure 3b](#)) with those in the stabilization phase ([Figure 4a,b](#)). While dendrites at the apex do not prefer synapses from any particular input group, in the stabilization phase, they acquire a preference for synapses from only two or three of the activity groups ([Figure 1b](#)). These dynamics resemble activity-dependent synaptic competition in the developing visual cortex, where asynchronously activated synapses tend to depress ([Winnubst et al., 2015](#)). In particular, the remaining synchronized synapses in our model experience correlation-dependent cooperation ([Kirchner and Gjorgjieva, 2021](#)), and therefore are able to stabilize the dendrite and prevent the total retraction of all branches.

This selective potentiation of synapses according to input correlation also leads to dendritic selectivity for inputs. In particular, synapses on the same dendrite are likely to be of the same activity group ([Figure 4c](#)). This selectivity is acquired throughout the simulation, where selectivity starts high (a nascent dendrite is trivially selective to its first synapse; $t = 0 - 1$ hours), drops almost to chance level (indiscriminate addition of synapses; $t = 9$ hours), and finally approaches a value of $\frac{1}{3}$ (two activity groups per dendrite remain after the pruning phase; $t = 72$ hours) ([Figure 4d](#)). To determine which activity group eventually stabilizes, we computed selectivity for each group early ($t = 9$ hours) and late ($t = 72$ hours). We found that early high (low) selectivity for an activity group translates into even higher (lower) selectivity in the stable state ([Figure 4e](#)), predicting an outsized impact of early synaptic contacts on continued dendritic function. Finally, we observed that when dendritic trees overlap strongly, they tend to be selective to different activity groups ([Figure 4f](#)) due to competition for limited potential synapses of any given group. Interestingly, also in the mouse visual cortex, neighboring neurons often exhibit different selectivity ([Ohki et al., 2005](#)), potentially reflecting the lasting impact of early competition between different inputs.

In summary, the emergence of dendrite selectivity for synapses of specific activity groups coincides with and supports the stabilization of dendritic morphologies.

Group activity correlations shape synaptic overshoot and selectivity competition across synaptic groups

Since correlations between synapses emerge from correlated patterns of spontaneous activity abundant during postnatal development ([Ackman et al., 2012](#); [Siegel et al., 2012](#)), we explored a wide range of within-group correlations in our model ([Figure 5a](#)). Although a change in correlations within the group has only a minor effect on the resulting dendritic lengths ([Figure 5b](#)) and overall dynamics, it can change the density of connected synapses and thus also affect the number of connected synapses to which each dendrite converges throughout the simulations ([Figure 5c,e](#)). This is due to the change in specific selectivity of each dendrite which is a result of the change in within-group correlations ([Figure 5d](#)). While it is easier for perfectly correlated activity groups to coexist within one dendrite ([Figure 5-Figure Supplement 1a](#), 100%), decreasing within-group correlations increases the competition between groups, producing dendrites that are selective for one specific activity group (60%, [Figure 5d](#), [Figure 5-Figure Supplement 1a](#)). This selectivity for a particular activity group is maximized at intermediate (approximately 60%) within-group correlations, while the contribution of the second most abundant group generally remains just above random chance levels ([Figure 5-Figure Supplement 1a](#)). Further reducing

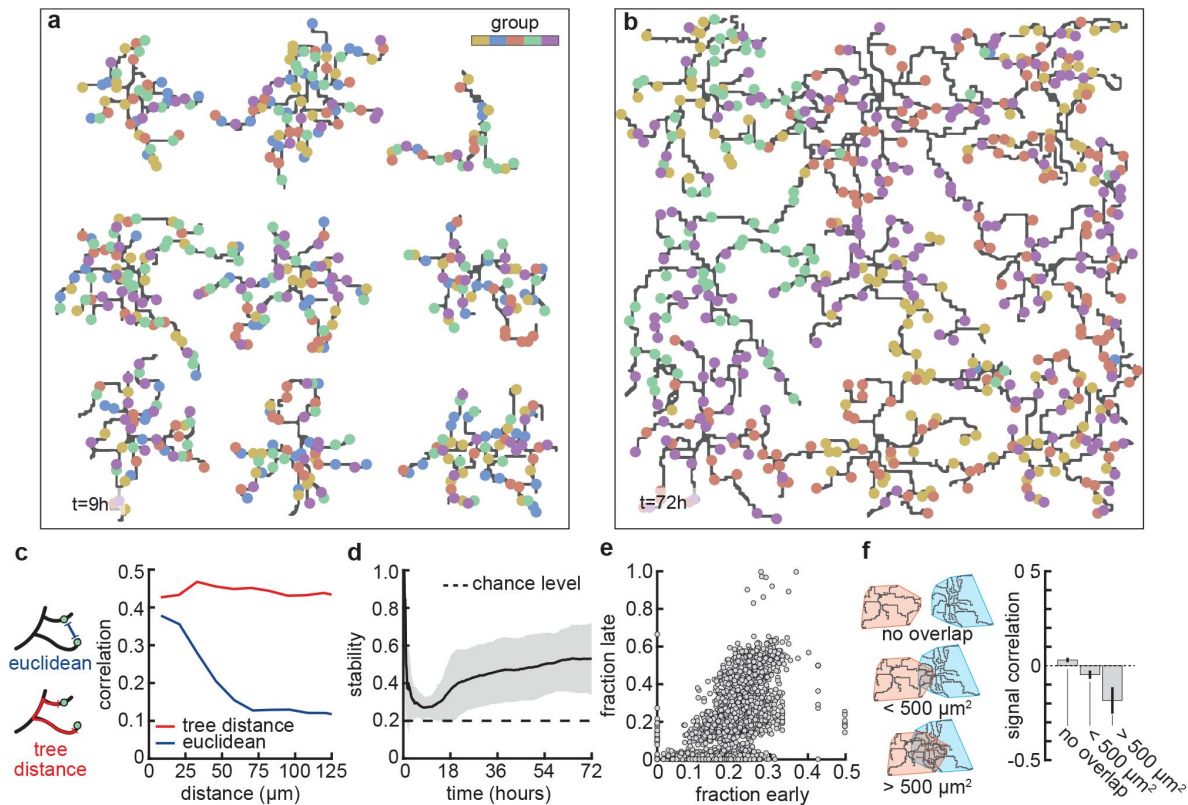


Figure 4.

Stable morphology is obtained through selective removal of synapses and dendritic input selectivity.

(a,b) Dendritic trees 72 hours) removal of synapses (Figure 3). Connected synapses colored corresponding to activity group, which Figure 1b. (c) Left: Schematic illustrating the difference between Euclidean and tree distance. Note that we between synapses from different trees. Right: Correlation between pairs of synapses as a function of the e distance (red). (d) Input selectivity of dendrites (defined as the fraction of the activity group with the highest time). Dashed line indicates chance level. All lines represent averages across 32 simulations with nine dendrites standard deviations. (e) Fraction of connected synapses per activity group early ($t=9$ hours) and late ($t=72$ hours) represents one of the five activity groups on one of the nine dendrites from the 32 simulations, resulting in Left: Schematic of different levels of overlap (rows) between the convex hulls of two dendrites, referring to the the dendrite. Right: Signal correlation (correlation between fractions of synapses from the same activity ndritic overlap. Error bars indicate the standard error of the mean, computed from 1152 pairs of dendrites from 32 simulations.

within-group correlations causes dendrites to lose their selectivity for specific activity groups due to the increased noise in the activity patterns (20%, **Figure 5a**). Overall, reducing within-group correlations increases synapse pruning (**Figure 5f**, bottom), also found experimentally (Matsumoto et al., 2024) as dendrites require an extended period to fine-tune connections aligned with their selectivity biases. This phenomenon accounts for the observed reduction in both the density and number of synapses connected to each dendrite.

In addition to the within-group correlations, developmental spontaneous activity patterns can also change correlations between groups as for example retinal waves propagated in different domains (Feller et al., 1997) (**Figure 5-Figure Supplement 2**). An increase in between-group correlations in our model intuitively decreases competition between the groups since fully correlated global events synchronize the activity of all groups (**Figure 5-Figure Supplement 2**). The reduction in competition reduces pruning in the model, which can be recovered by combining cross-group correlations with decreased within-group correlations (**Figure 5-Figure Supplement 2**). Our simulations show that altering the correlations within activity groups increases competition (by lowering the within-group correlations) or decreases competition (by raising the across-group correlations). Hence, in our model, competition between activity groups due to non-trivially structured correlations is necessary to generate realistic dynamics between activity-independent growth and activity-dependent refinement or pruning.

In sum, our simulations demonstrate that our model can operate under various correlations in the spike trains. We find that the level of competition between synaptic groups is crucial for the activity-dependent mechanisms to either potentiate or depress synapses and is fully consistent with recent experimental evidence showing that the correlation between spontaneous activity in retinal ganglion cells axons and retinal waves in the superior colliculus governs branch addition vs. elimination (Matsumoto et al., 2024).

Balance of mature and immature brain-derived neurotrophic factors controls the arborization of dendrites

After establishing that our model can capture important aspects of the dynamics of dendritic development through the combination of activity-independent and activity-dependent mechanisms, including local plasticity, we asked how changing properties of the plasticity rule might affect dendritic growth and synaptic organization. Developmentally, the interaction between two neurotrophic factors, BDNF and proBDNF (**Figure 6a**), has been found to play a key role in the organization of synaptic inputs into clusters (Niculescu et al., 2018). Therefore, through the previously established link between this neurotrophin interaction and synaptic plasticity (Kirchner and Gjorgjieva, 2021), we investigated the influence of changing the underlying molecular interactions on dendritic morphology.

As we have previously shown, the ‘offset’ term in our plasticity rule (Equation 1) represents the neurotrophin balance (computed as $\text{BDNF}/(\text{BDNF} + \text{proBDNF})$) released upon stimulation of a synapse (Kirchner and Gjorgjieva, 2021). Consequently, we found that an overabundance of BDNF (proBDNF) leads to potentiation (depression) of the synapse (**Figure 6b**), consistent with experimental data (Lu et al., 2005). Furthermore, our plasticity rule acts locally on the dendrite, so that the strength of individual synapses is affected by interactions with other nearby synapses. Concretely, a lower (higher) density of nearby synapses tends to lead to potentiation (depression) of the synapse (Kirchner and Gjorgjieva, 2021).

To better understand the interactions between the balance of neurotrophins and the density of synapses, we analytically derived the maximum density of synapses that can stabilize given a balance of neurotrophins (**Figure 6c**, see Methods). We found that an overabundance of BDNF (proBDNF) leads to a higher (lower) maximal density of synapses (**Figure 6c**). Indeed, when we simulated dendritic development with varying neurotrophin ratios, we found that the density of

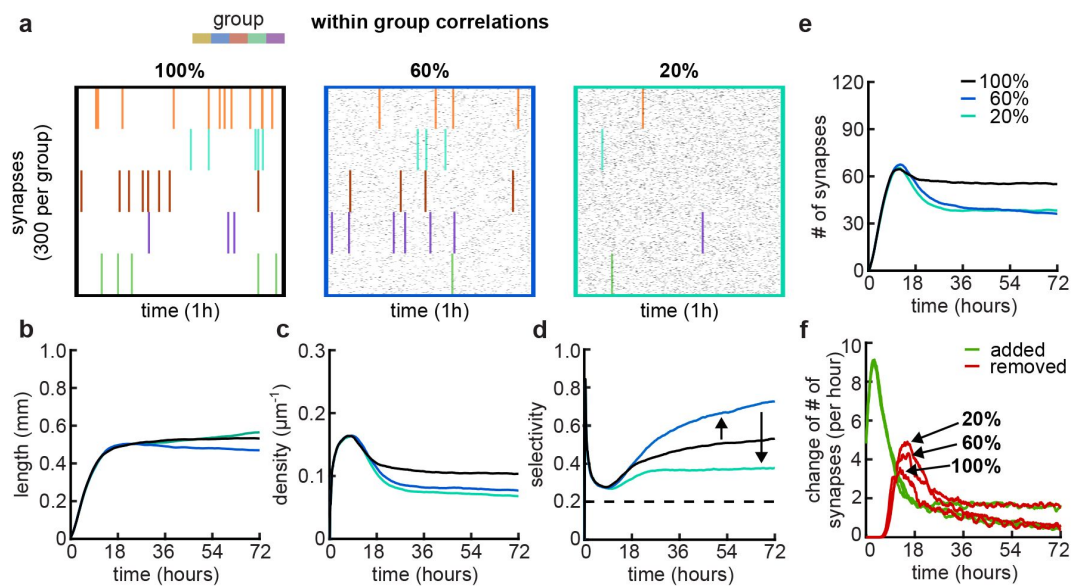


Figure 5.

Effects of within-group correlations on group selectivity and synaptic pruning dynamics.

(a) Correlated spike train examples of all the activity groups with different within-group correlation statistics (from left to right, 100%, 80% and 20% within-group correlations). Spike trains are sorted by synapses that correspond to each of the activity groups. Correlated spikes that are shared across all the synapses of one group are highlighted by the group's color. (b) Total dendritic length as a function of time. (c) Total density of synapses on the dendrite. (d) Input selectivity of dendrites (defined as the fraction of the activity group with the highest representation) as a function of time. Dashed line indicates chance level. (e) Total number of connected synapses as a function of time. (f) Added (green) and pruned synapses (red) as a function of time. Arrows indicate the peaks of the simulations with different correlations. All lines represent averages across 32 simulations with nine dendrites each.

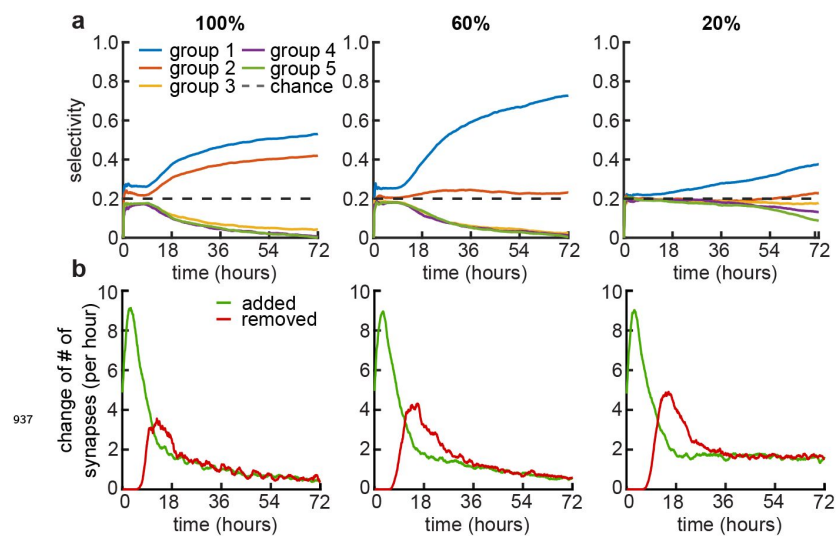


Figure 5-Figure supplement 1.

Examples of coexisting activity groups and synaptic pruning with low within-group correlations.

(a) Average selectivity of the simulated dendrites for the five activity groups (groups are sorted by their contribution to the dendrites synapses, where group 1 always has the most synapses contributing and group 5 the least) for three within-group correlations 100% (left), 60% (middle) and 20% (right). Dashed line indicates chance level. (b) Change in the number of synapses for three within-group correlations 100% (left), 60% (middle) and 20% (right).

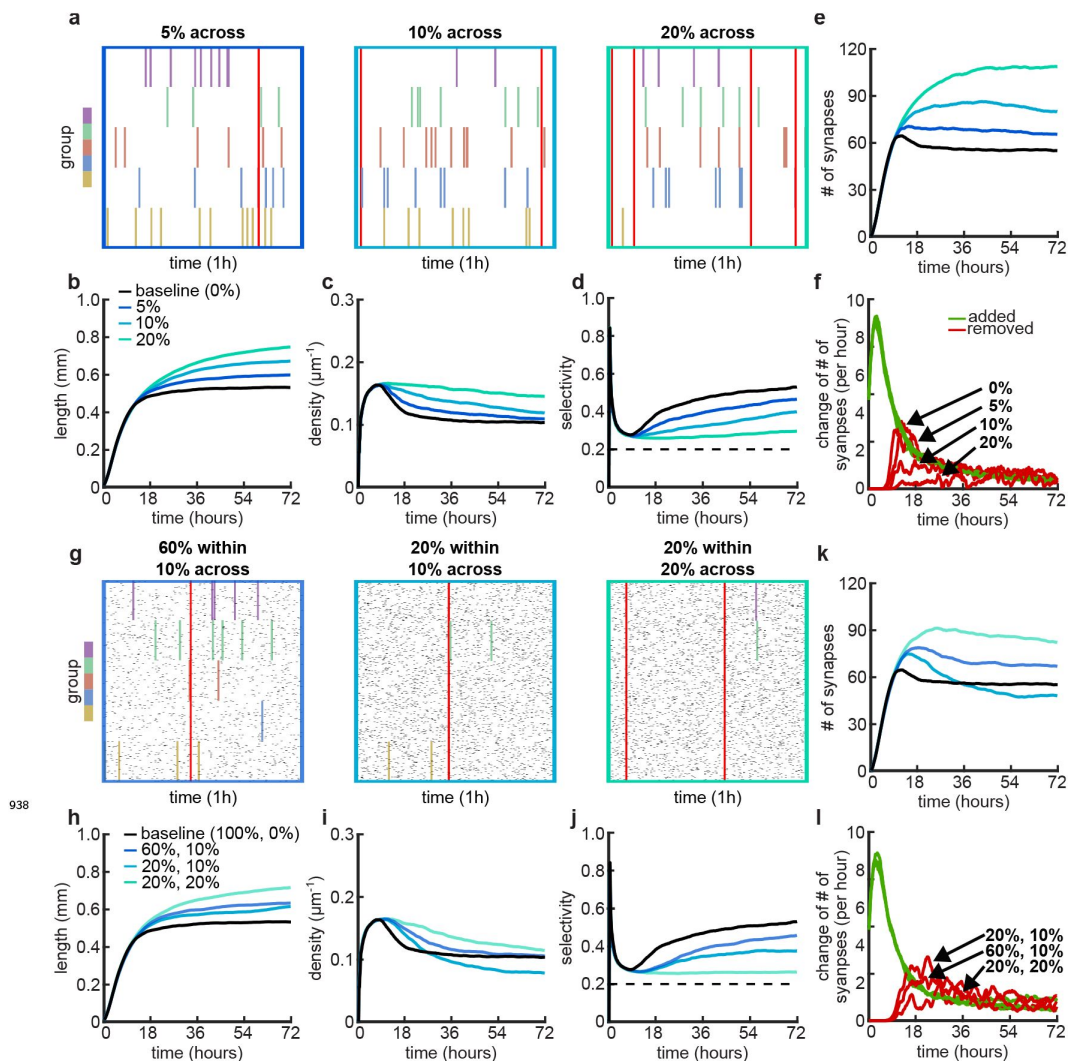


Figure 5-Figure supplement 2.

Effects of across-group correlations and combinations of within- and across-group correlations on group selectivity and synaptic pruning dynamics. (a,g) Correlated spike train examples of all the activity groups with different across-group correlations (a) or combinations of within- and across-group correlations (f). Spike trains are sorted by synapses that correspond to each of the activity groups. Correlated spikes that are shared across all the synapses of one group are highlighted by the group's color, and global spike events that are shared across all synapses of all groups are shown in red. (b,h) Total dendritic length as a function of time. (c,i) Total density of synapses on the dendrite. (d,j) Input selectivity of dendrites (defined as the fraction of the activity group with the highest representation) as a function of time. Dashed line indicates chance level. (e,k) Total number of connected synapses as a function of time. (f,l) Added (green) and pruned synapses (red) as a function of time. Arrows indicate the peaks of the simulations with different correlations. All lines represent averages across 32 simulations with nine dendrites each.

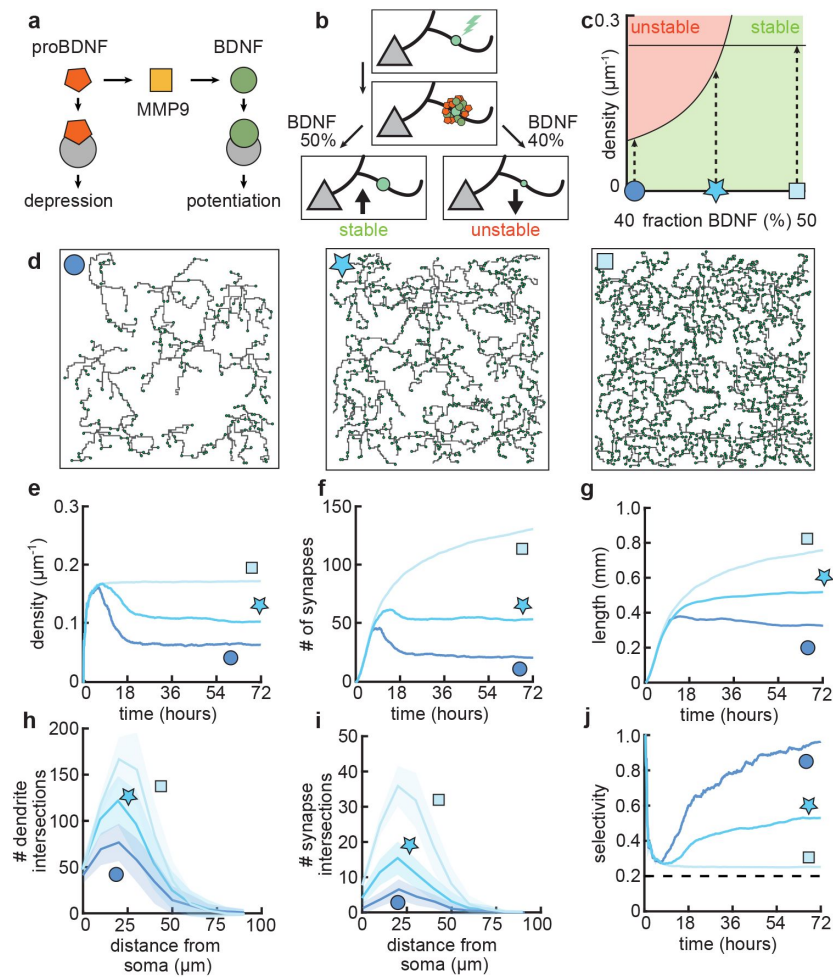


Figure 6.

Dendritic arborization is controlled by the ratio of neurotrophic factors.

(a) Interactions between molecular factors underlie a local activity-dependent plasticity rule for synaptic change (Equation 1, (Kirchner and Gjorgjieva, 2021)). Neurotrophins (BDNF and proBDNF) bind to different neurotrophin receptors, and a cleaving protease (MMP9) converts proBDNF to BDNF in an activity-dependent manner. (b) Schematic illustrating the impact of different concentrations of BDNF on synaptic change. Upon stimulation of a synapse (top), proBDNF and BDNF is released into extracellular space (middle), where proBDNF can be cleaved into BDNF by MMP9. Depending on the neurotrophin ratio, computed as $\text{BDNF}/(\text{BDNF} + \text{proBDNF})$, the synapse is stabilized (left) or depressed and hence eventually removed (right). (c) Maximally possible stable density of synapses as a function of the initial concentration of BDNF. Stable (no pruning; green) and unstable (pruning occurs; red) areas are indicated. (d) Three examples of dendrites with superimposed synapses (green) with high initial BDNF concentration (49%), the baseline concentration (46.5%, same as Figure 1–Figure 3) and low initial BDNF (43%). Symbols correspond to locations marked in panel c. (e–g) Averages for density of synapses on the dendrite (e), number of connected synapses (f) and total length of dendrite (g) as a function of time for dendrites from the three conditions shown in d. (h–i) Average number of dendrite intersections (h) and synapses (i) as a function of distance from the soma for dendrites from the three conditions shown in d. (j) Global selectivity as a function of time for dendrites from the three conditions shown in d. Dashed line indicates chance level. All lines represent averages across 32 simulations with nine dendrites each. Shaded area indicates two standard deviations.

synapses per dendrite increases with increasing neurotrophin ratio (**Figure 6d,e**). Consistent with biological evidence (McAllister et al., 1995; Tyler and Pozzo-Miller, 2001), in our model, developing dendrites treated with BDNF tend to grow larger and have a higher density of synapses (**Figure 6e,g**). In contrast, over-expression of proBDNF leads to smaller dendrites with fewer synapses (Koshimizu et al., 2009; Yang et al., 2014) (**Figure 6f,g**). Perturbing the balance between neurotrophins scales the Sholl diagram of dendrite intersections and synapses, but does not qualitatively affect the shape of the curve (**Figure 6h,i**).

In our model, these changes in length and density are explained by a change in the selectivity of the synapses (**Figure 6j**). Concretely, an increase in BDNF erases all synaptic competition, reducing the selectivity to chance level, while an increase in proBDNF greatly amplifies synaptic competition and thus selectivity. These differences in competition determine the number of pruned synapses and thus the length at which the dendrite stabilizes. Thus, our model predicts that biologically-relevant differences in dendritic morphology may arise from different neurotrophin ratios due to the maximal density of synapses that stabilizes the dendrite.

Different impacts of activity-dependent and -independent factors on dendritic development

Our mechanistic model enabled us to dissect the different roles of activity-dependent and -independent mechanisms on dendritic morphology. To this end, we varied only activity-dependent factors or only activity-independent factors across a set of simulations (**Figure 7a**). We introduced variability in the activity-dependent aspects of the model through the firing patterns of potential synapses, and in the activity-independent aspects of the model via fluctuations in both the extrinsic growth signals and the intrinsic mechanisms underlying dendrite growth (see Methods, **Figure 7b**).

Consistent with experiments (Scala et al., 2021), dendrites produced by our model exhibit substantial variability in morphology (**Figure 7a**), length (**Figure 7c**), and number of synapses (**Figure 7d**). Comparing dendrites that experienced either identical activity-dependent or identical activity-independent factors allowed us to compute the percentage of change in morphology attributable to each factor as a function of developmental time (**Figure 7e,f**). We found that while activity-independent factors tend to lead to large differences in morphology early on, activity-dependent factors affect dendrite morphology with a substantial delay. These differences can be explained by the delay in synaptic pruning relative to initial synaptic formation (**Figure 3d**).

Despite substantial variability, there are predictive factors for the final length of the dendrite. In particular, we found a positive relationship between the number of major branches, i.e. branches starting from the soma, and the final length (**Figure 7g**). Interestingly, this is consistent with dendrites reconstructed from multiple regions of the mouse cortex (**Figure 7–Figure Supplement 1**). Furthermore, our model predicts that dendrites that have a high (low) total length early on will, on average, retain a (high) low total length throughout development (**Figure 7e**).

Thus, our model suggests that while activity-independent factors affect dendritic morphology early on during development, activity-dependent factors dominate later. Properties such as the number of major branches or the length of dendrites during early development might be predictive of the dendrite's morphology throughout the animal's lifetime.

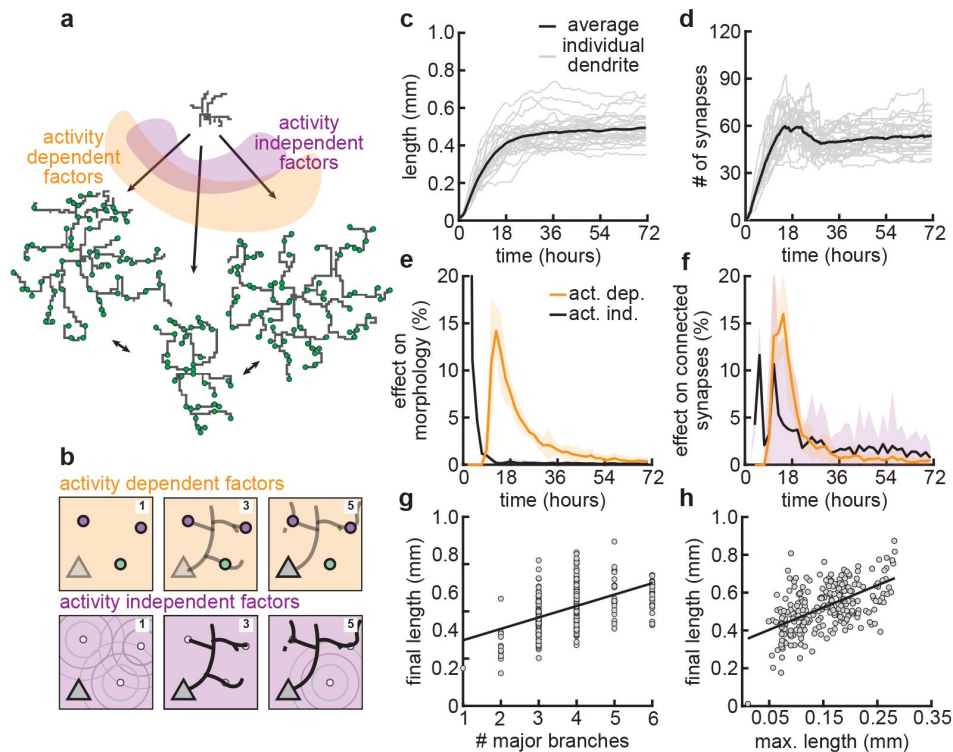


Figure 7.

Morphological variability emerges from the interaction of activity-dependent and -independent factors.

(a) Example of three dendrites with identical initial conditions but different random seeds. The colors illustrate that initial growth is governed by activity-independent factors, while later growth is governed by activity-dependent factors. (b) Schematic illustrating how variability is introduced into model: activity-dependent via the patterns of spontaneous activity (orange), and activity-independent via fluctuations in both the extrinsic growth stimulating field (purple 1) and the intrinsic mechanisms underlying dendrite growth (purple 2; see Methods). (c,d) Total length (c) and number of synapses (d) as a function of time for dendrites with identical initial conditions but different random seeds. Each gray line corresponds to one dendrite from one of 32 simulations. Bold line represents average. (e,f) Percentage in change of morphological similarity (e) and similarity of connected synapses (f) as a function of time for simulations where activity-dependent (orange) or -independent (purple) factors vary. Lines represent averages across 32 simulations with nine dendrites each. Shaded area indicates two standard deviations. (g,h) Final length as a function of number of major branches (g) and maximal length in the first 18 hours of the simulation (h). Lines indicate linear regression.

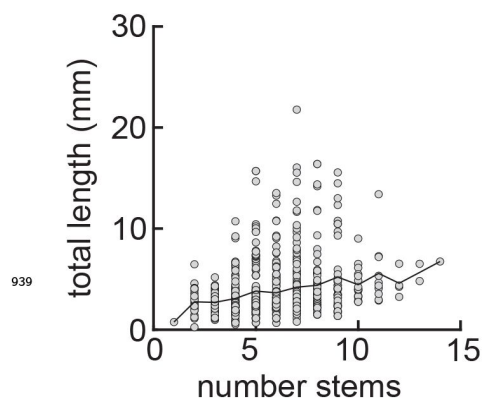


Figure 7–Figure supplement 1.

Total tree length increases with the number of stems.

Average (solid black) and individual (gray circles) total tree lengths as a function of the number of stems. $N = 701$. Data from the Allen Cell Types Database (2015).

Coupled dendrite growth and synapse formation leads to approximately optimal minimal wiring

Given the spatial constraints within cortical architecture and the significant metabolic costs associated with sustaining complex dendritic morphologies, it is advantageous for neurons to establish synaptic connections using the shortest possible dendritic length (Cuntz et al., 2012 [↗](#)). (Figure 8a [↗](#), top vs. middle). Yet another constraint that neurons must adhere to as brain size increases is the need to minimize conduction delays. (Figure 8a [↗](#), bottom). A balance between these two conservation laws constrains dendrite morphology (Cuntz et al., 2010 [↗](#), 2012 [↗](#); Ferreira Castro et al., 2020 [↗](#)). In particular, in our model, dendrites are assumed to grow towards the nearest potential synapse. Thus, we investigated how the final dendritic length produced by our models compares to the principles of optimal wiring.

As an initial analysis, our aim was to assess the upper and lower bounds to which our models could be constrained. As a lower limit for wiring minimization, we used known scaling relationships where the length (L) of a minimally wired dendrite in a plane scales with the square root of the number of synapses (N) times the area over which the synapses are distributed (A): $L = \sqrt{NA/\pi}$ (Cuntz et al., 2012 [↗](#)). In contrast, as an upper bound, the length of a dendrite when synapses are connected randomly scales with the number of connected synapses times the average distance of two random points on a circle (Uspensky, 1937 [↗](#)), $L = N^{1/3} \sqrt{A/(2\pi)}$. This differs from the optimal wiring solution by a square root in the number of synapses. Using the convex hull that circumscribes the stabilized synapses as the area over which the synapses are distributed (Figure 8b [↗](#)), we compared the actual length of the dendrite with minimal and the random wiring length (Figure 8c [↗](#)). Our analysis reveals that the dendritic lengths generated by our simulations show values that are shorter compared to those predicted by random wiring models, yet they are longer than those predicted by the theoretical minimum length.

Next, we wanted to know if the deviation from minimal wiring might quantitatively match the one observed in real dendrites. To investigate this question, we reanalyzed a published dataset (Cuntz et al., 2012 [↗](#)) containing the total lengths and the number of branch points of 13,112 dendrites pooled across 74 sources. When computing the fold change between the real and the theoretical expectations derived from the scaling law in the dataset, we confirmed that real dendrites tend to be consistently larger than the theoretical minimum (Figure 8d [↗](#)). Interestingly, the fold change between the length of real dendrites and the theoretical minimum is similar to the fold change of our simulated dendrites and the theoretical minimum (Figure 8e [↗](#)). This deviation is expected given that real dendrites need to balance their growth processes between minimizing wire while reducing conduction delays. The interplay between these two factors emerges from the need to reduce conduction delays, which requires a direct path length from a given synapse to the soma, consequently increasing the total length of the dendritic cable. (Cuntz et al., 2010 [↗](#), 2012 [↗](#); Ferreira Castro et al., 2020 [↗](#)).

To investigate this further, we compared the scaling relations of the final morphologies of our models with other synthetic dendritic morphologies generated using a previously described minimum spanning tree (MST) based model. The MST model balances the minimization of total dendritic length and the minimization of conduction delays between synapses and the soma. This balance results in deviations from the theoretical minimum length because direct paths to synapses often require longer dendrites (Cuntz et al., 2008 [↗](#), 2010 [↗](#)). The balance in the model is modulated by a balancing factor (bf). If bf is zero, dendritic trees minimize the cable only, and if bf is one, they will try to minimize the conduction delays as much as possible. It is important to note that the MST model does not simulate the developmental process of dendritic growth; it is a phenomenological model designed to generate static morphologies that resemble real cells.

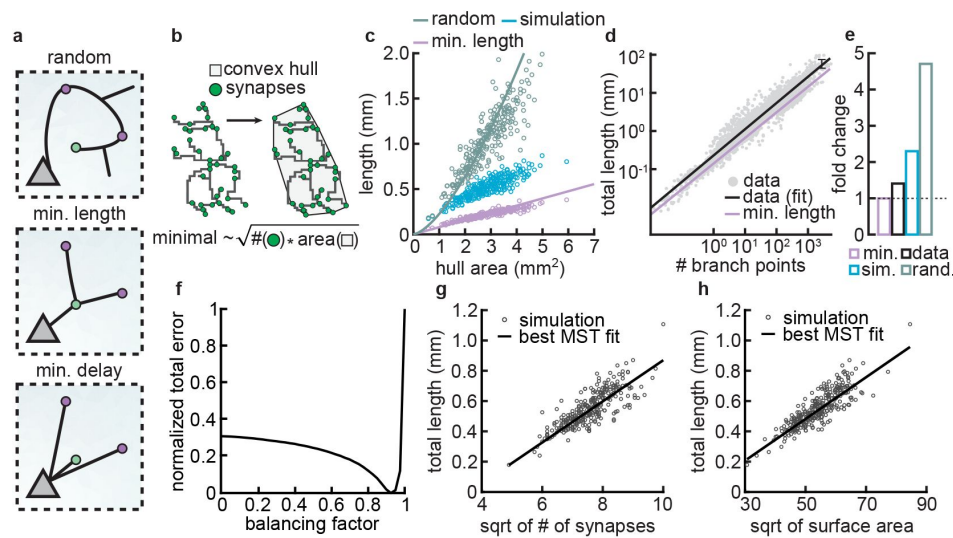


Figure 8.

Synthetic dendritic morphologies: Analysis of optimal wiring principles.

(a) Schematic of synaptic wiring principles: random (top), minimal length (middle), and minimal delay (bottom) for a set of synapses. (b) The convex hull of all synapses connected to a dendrite with the proportionality of minimal wiring length (bottom). (c) Total tree length as a function of convex hull area in the minimal wiring length, simulated and random scenario. Each dot corresponds to one of 288 dendrites from 32 simulations. Lines correspond to analytic predictions for the average density across all simulations. (d) Total tree length against the number of branch points in log scale, both for data and theoretical minimum. Data extracted from (Cuntz et al., 2012). (e) Total tree length in the data (black, average of $n=13,112$), our simulations (blue, average of 288 dendrites from 32 simulations), and the random baseline (green, analytically computed) relative to the theoretical minimum (pink, analytically computed). (f) Model parameter bf vs. error (comparing total length, number of terminals and surface area, with corresponding synthetic trees (straight black line; optimal $bf = 0.9250$). (g-h) Scaling behaviour of the square root of number of synapses against total length (g, $R^2 = 0.65$) and square root of surface area against total length (h, $R^2 = 0.73$) showing the relationships expected from the optimal wire equations. The black line shows the best MST bf fit scaling behaviour (see Methods). In both panels, each dot represents one synthetic tree ($n=288$). Lines were fit using linear regression.

To facilitate the comparison of total lengths between our simulated and MST morphologies, we generated MST models under the same initial conditions (synaptic spatial distribution) as our models and simulated them to match several morphometrics (total length, number of terminals, and surface area) of our grown morphologies. This allowed us to create a corresponding MST tree for each of our synthetic trees. Consequently, we could evaluate whether the branching structures of our models were accurately predicted by minimum spanning trees based on optimal wiring constraints. We found that the best match occurred with a trade-off parameter $bf = 0.9250$ (**Figure 8f**). Using the morphologies generated by the MST model with the specified trade-off parameter (bf), we showed that the square root of the synapse count and the total length (L) in both our model generated trees and the MST trees exhibit a linear scaling relationship (**Figure 8g**; $R^2 = 0.65$). The same linear relationship can be observed for the square root of the surface area and the total length L of our model trees and the MST trees (**Figure 8h**; $R^2 = 0.73$). Overall, these results indicate that our model generate trees are well-fitted by the MST model and follow wire optimization constraints.

We acknowledge that the value of the balancing factor bf in our model is higher than the range of balancing factors that is typically observed in the biological dendritic counterparts, which generally ranges between 0.2 and 0.4 (Cuntz et al., 2012; Ferreira Castro et al., 2020; Baltruschat et al., 2020). However, it is still remarkable that our model, which does not explicitly address these two conservation laws, achieves approximately optimal wiring. Why do we observe such a high bf value? We reason that two factors may contribute to this. First, in our models, local branches grow directly to the nearest potential synapse, potentially taking longer routes instead of optimally branching to minimize wiring length (Wen and Chklovskii, 2008). Second, the growth process in our models does not explicitly address the tortuosity of the branches, which can increase the total length of the branches used to connect synapses. In the future, it will be interesting to add constraints that take these factors into account. Taken together, combining activity-independent and -dependent dendrite growth produces morphologies that approximate optimal wiring.

Discussion

Dendritic growth and the formation, stabilization, and removal of synapses during early development depend on various factors, including extrinsic factors such as growth cues, intrinsic molecular signaling, and correlated patterns of spontaneous activity. However, the nature of these interactions and the implications for dendritic function throughout life remain largely unexplored. In this study, we proposed a mechanistic model for the growth and retraction of dendritic branches, as well as the formation and removal of synapses on these dendrites during development, based on the interaction of activity-independent cues from potential synaptic partners and local activity-dependent synaptic plasticity. Our model can simultaneously capture two main aspects of dendritic growth: producing dendritic morphologies and driving synaptic organization.

Assumptions and predictions of the model

Some of the most prominent models of dendritic growth have focused on activity-independent rules based on geometric or biophysical constraints (Cuntz et al., 2010, 2012). Despite their immense success in generating realistic dendritic morphologies, they leave open the question of the underlying biological mechanisms. Other studies have implemented global activity-dependent rules that require feedback signals from the soma or the axon (Ooyen et al., 1995). Our model proposes a simple and biologically plausible mechanism for the simultaneous dendritic growth and synaptic organization based on activity-independent cues and local activity-dependent learning rules, which cluster synaptic inputs according to functional preferences. Numerous

experimental studies have demonstrated the importance of such local plasticity for the emergence of local synaptic organization in the form of clusters, as well as dendritic function (Hering and Sheng, 2001 [↗](#); Lohmann et al., 2002 [↗](#); Chen et al., 2013 [↗](#); Niculescu et al., 2018 [↗](#)).

Our model makes some simplifying assumptions at the expense of mechanistic insights. For instance, we model the generation of only stellate-like morphologies without the apical trunk. Many types of neurons are characterized by stellate morphologies, especially in the somatosensory cortex (Schubert et al., 2003 [↗](#); Marques-Smith et al., 2016 [↗](#); Scala et al., 2019 [↗](#)). Nonetheless, it would be interesting to investigate if our model's mechanisms can be minimally modified to apply to the generation of apical dendrites. Moreover, we generate our model dendrites in a two-dimensional, flat sheet of cortex. We anticipate that the models can be straightforwardly extended to three dimensions, but with additional computational cost. Although our assumptions may be too simplified to generate perfectly biologically realistic morphologies, the simple rules in our model capture basic morphological features, such as the number of branches, the total length, and the Sholl analysis, with those of biological neurons reported in the literature.

A key advantage of our mechanistic model is the ability to predict the impact of early perturbations on mature dendritic morphology, as the model allows us to independently investigate activity-independent and -dependent influences on dendrite growth and synaptic organization. For example, three distinct phases of synapse development, overshoot, pruning, stabilization, and stable dendritic trees, emerge naturally from the interactions between activity-independent signaling and activity-dependent synaptic plasticity, without additional assumptions and are robust to parameter variations. The stabilization of dendritic morphologies in our model is enabled by the emergence of input selectivity, which implies local organization of synapses responsive to a particular correlated input pattern on the dendrite. Hence, our model explains how dendritic morphology can adapt to changes in activity-dependent plasticity or input statistics during development, as observed experimentally (Cline and Haas, 2008 [↗](#); McAllister et al., 1995 [↗](#); Tyler and Pozzo-Miller, 2001 [↗](#); Matsumoto et al., 2024 [↗](#)). Further, we provide a mechanistic explanation for the emergence of approximately optimal minimal wiring in cortical dendrites. Thus, our model provides a new perspective on the interaction of activity-independent and -dependent factors influencing dendrite growth and suggests that the formation and activity-dependent stabilization vs. removal of synapses might exert powerful control over the growth process.

The correlated activity experienced by our modeled synapses (and resulting synaptic organization) does not necessarily correspond to visual orientation, or any stimulus feature, for that matter, but is rather a property of spontaneous activity. Nonetheless, there is some variability in what the experimental data show. Many have shown that synapses on dendrites are organized into functional synaptic clusters: across brain regions, developmental ages and diverse species from rodent to primate (Kleindienst et al., 2011 [↗](#); Winnubst et al., 2015 [↗](#); Iacaruso et al., 2017 [↗](#); Scholl et al., 2017 [↗](#); Niculescu et al., 2018 [↗](#); Takahashi et al., 2012 [↗](#); Gökçe et al., 2016 [↗](#); Wilson et al., 2016 [↗](#); Kerlin et al., 2019 [↗](#); Ju et al., 2020 [↗](#); Hedrick et al., 2022 [↗](#), 2024 [↗](#)). Other studies have reported lack of fine-scale synaptic organization (Chen et al., 2013 [↗](#); Varga et al., 2011 [↗](#); Chen et al., 2011 [↗](#); Jia et al., 2010 [↗](#), 2014 [↗](#)). Interestingly, some of these discrepancies might be explained by different species showing clustering with respect to different stimulus features (orientation or receptive field overlap) (Scholl et al., 2017 [↗](#); Wilson et al., 2016 [↗](#); Iacaruso et al., 2017 [↗](#)). Our prior work proposed a theoretical framework to reconcile these data: combining activity-dependent plasticity as we used in the current work, and a receptive field model for the different species (Kirchner and Gjorgjieva, 2021 [↗](#)).

Comparison with the synaptotrophic hypothesis

The synaptotrophic hypothesis, originally proposed three decades ago (Vaughn, 1989 [↗](#)), has provided a useful framework for interpreting the effect of neural activity and synapse formation on dendrite development. Our proposed model is inspired by the synaptotrophic hypothesis, but

differs from it in a few key aspects. (1) The synaptotrophic hypothesis postulates that synaptic activity is necessary for dendrite development (Cline and Haas, 2008 [↗](#)). In contrast, our model contains an activity-independent component that allows dendrites to grow even in the absence of synaptic activity. Our model is thus consistent with the finding that even in the absence of neurotransmitter secretion connected neuronal circuits with morphologically defined synapses can still be formed (Verhage et al., 2000 [↗](#); Ferreira Castro et al., 2023) and with computational (non-mechanistic) models that produce dendrites with many relevant morphological properties without relying on activity (Cuntz, 2016 [↗](#)). (2) The synaptotrophic hypothesis does not specify the exact molecular factors underlying the information exchange pre- and postsynaptically. Based on recent experiments that identify central molecular candidates (Winnubst et al., 2015 [↗](#); Kleindienst et al., 2011 [↗](#); Niculescu et al., 2018 [↗](#); Lu et al., 2005 [↗](#)), our model proposes a concrete mechanistic implementation based on neurotrophic factors (Kirchner and Gjorgjieva, 2021 [↗](#)). (3) The synaptotrophic hypothesis postulates that whether a potential synaptic contact is appropriate can be rapidly evaluated pre- and postsynaptically. Inspired by experiments (Lohmann et al., 2002 [↗](#); Niell et al., 2004 [↗](#)), the fate of a synapse in our model is determined only within tens of minutes or hours after it is formed. This is due to the slow timescale of synaptic plasticity.

Although we did not explicitly model postsynaptic firing, our previous work with static dendrites has shown that it can play an important role in establishing a global organization of synapses on the entire dendritic tree of the neuron (Kirchner and Gjorgjieva, 2021 [↗](#)). For example, we showed that it could lead to the emergence of retinotopic maps on the dendritic tree which have been found experimentally (Iacarus et al., 2017 [↗](#)). Since we use the same activity-dependent plasticity model in this paper, we expect that the somatic firing will have the same effect on establishing synaptic distributions on the entire dendritic tree.

Relationship between dendritic form and function

In contrast to previous studies which focused on how dendritic morphology affects function, e.g. through nonlinear signal transformation (Poirazi and Papoutsis, 2020 [↗](#)) or dynamic routing of signals (Payeur et al., 2019 [↗](#)), we propose that dendrite form and function reciprocally shape each other during development. While the morphology of the dendrite constrains the pool of available potential synapses, synaptic activity determines the dendritic branch's stability (Fig. 1 [↗](#)). As a result, the dendritic tree self-organizes in an appropriate shape to support a limited number of functionally related synapses. These initial synaptic contacts might then serve as a scaffold around which additional functionally related synapses cluster to form the building blocks to support the powerful computations of mature dendrites (Kirchner and Gjorgjieva, 2022 [↗](#)).

Dynamics of dendritic development

Here, we focus on the early developmental period of highly dynamic dendritic growth and retraction. However, dendritic morphology remains remarkably stable in later development and through-out adulthood (Richards et al., 2020 [↗](#); Ferreira Castro et al., 2020 [↗](#); Koleske, 2013 [↗](#)). This stability is achieved despite substantial increases in the overall size of the animal (Richards et al., 2020 [↗](#); Ferreira Castro et al., 2020 [↗](#)) and ongoing functional and structural plasticity of synapses (Kleindienst et al., 2011 [↗](#); Winnubst et al., 2015 [↗](#); Kirchner and Gjorgjieva, 2021 [↗](#)). Although it is still unclear how exactly synaptic organization is established during early development and how synapses are affected by the overall increase in dendrite size, somatic voltage responses to synaptic activity are largely independent of dendrite size (Cuntz et al., 2021 [↗](#)). The stability of dendrites has been shown to play a crucial role in allowing proper function of the adult brain and is affected in many psychiatric disorders and neurodegenerative diseases. In particular, the release of BDNF, which is related to synaptic activity, affects structural consolidation of dendrites and therefore long-term stability (Koleske, 2013 [↗](#)). Our mechanistic model allows us to perturb the balance of neurotrophic factors and investigate the effects on dendritic development. For instance, our model predicts detrimental effects on dendrite stability

as a result of extreme or non-existent input selectivity, providing insights into functional consequences of disrupted dendrite growth in neurodevelopmental disorders (Johnston et al., 2016 [↗](#)).

Interneurons and inhibitory synapses

In addition to excitatory neurons and synapses that are the focus of this study, inhibitory interneurons and inhibitory synapses also play an important role in brain development (Naskar et al., 2019 [↗](#)). Interneurons fall into genetically distinct subtypes, which tend to target different portions of pyramidal neurons (Rudy et al., 2011 [↗](#); Kepecs and Fishell, 2014 [↗](#)). In particular, somatostatin-expressing interneurons (SST) preferentially target the dendrites of pyramidal neurons, while parvalbumin-expressing interneurons (PV) preferentially target the soma. Furthermore, the dendrites of inhibitory neurons have a complex morphology that likely allows them to perform intricate transformations of incoming signals (Tzilivaki et al., 2019 [↗](#), 2021 [↗](#)). Investigating whether and how inhibitory interneurons and synapses might interact with excitatory ones during dendritic development is an exciting direction for future research.

In summary, by proposing a mechanistic model of dendritic development that combines activity-independent and activity-dependent components, our study explains several experimental findings and makes predictions about the factors underlying variable dendritic morphologies and synaptic organization. Interestingly, the stable morphologies it generates are approximately optimal in terms of wiring length and experimental data. Finally, our model provides the basis for future exploration of different learning rules and cell types which could differ across brain regions, species, and healthy vs. disease states.

Methods and materials

Activity-independent synaptic signals

In the synaptotrophic hypothesis, dendrite growth is directed towards potential synaptic partners. In our model, we captured this aspect by introducing a *growth field* of activity-independent synaptic signals, $T(\mathbf{p})$, over all positions $\mathbf{p}$ in our sheet of cortex. This field contains point sources at the positions of potential synapses, $\mathbf{p}_i$, and evolves over time according to a diffusion equation,

$$T(\mathbf{p})^{t+1} = T(\mathbf{p})^t \circ D + \mu \sum_i \mathbf{p}_i + \sigma \mathbf{N}. \quad (2)$$

The growth field at time point $t+1$ is therefore given by the sum of the growth field at time t convolved with a diffusion filter D , a constant input of size μ from all potential synapses, which are currently not connected to a dendrite, as well as independent Gaussian noise, $\mathbf{N}$, with standard deviation μ . We chose a two dimensional Gaussian for the diffusion filter D , making the field $T(\mathbf{p})$ mathematically equivalent to a heat diffusion in two dimensions (**Figure 1–Figure Supplement 1** [↗](#)).

Asynchronous dendrite growth and retraction

Dendrite development critically depends on resources from the soma (Ye et al., 2007 [↗](#)). Consequently, we modeled the growth of dendrites to depend on *scouting agents* that spread outward from the soma at regular time intervals, v_{scout} , and that traverse the dendritic tree at speed t_{scout} (**Figure 1–Figure Supplement 2** [↗](#)). These scouting agents resemble actin-blobs that control dendrite growth (Nithianandam and Chien, 2018 [↗](#)). When a scouting agent encounters a branch point, there is a 0.5 chance for it to continue in any direction. This means it can go in one direction, but it can also duplicate or disappear completely. We further modeled these scouting agents to detect the growth field's value – a scouting agent stops at a position on the dendrite

where this field is locally maximal and marks this location for growth. The dendrite will then expand at the marked positions in the direction of the gradient of the growth field, and the scouting agent moves to this new position. If the dendrite grows to the location of a potential synapse, this synapse is then realized, and its initial weight is set to $w_{init} = \frac{1}{2}$. Two branches of the same neuron may never become adjacent; however, branches from other neurons may be crossed freely. If a scouting agent reaches the end of a branch without finding a local maximum of the growth field along its path, the scouting agent initiates the retraction of this branch. Depending on proximity, a branch then retracts up to the nearest stable synapse, the next branch point, or the soma. Because our simulations are a spatially discrete approximation of a biological flat sheet of cortex, we had to ensure that growth behaves appropriately in cases where the discretization scheme becomes relevant (**Figure 1–Figure Supplement 2** [↗](#)).

Minimal plasticity model

When a synapse k forms on the dendrite, its weight w_k evolves according to a previously proposed minimal plasticity model for interactions between synapses on developing dendrites (Kirchner and Gjorgjieva, 2021 [↗](#)). This model can be linked to a full neurotrophin model that interprets the parameters in terms of the neurotrophic factors BDNF, proBDNF, and the protease MMP9. In this model, the k -th synapse is stimulated within input event trains x_k

$$x_k(t) = \int_0^\infty \sum_j \delta(s - s_k^j) (H(t - s) - H(t - x_{dur} - s)) ds \quad (3)$$

with events at times t_k^j and where the Heaviside step function $H(t)$ is 0 when t is less than 0 and 1 when t is greater or equal than 0, so that events have duration x_{dur} (50 time steps). The minimal plasticity model consists of a synapse-specific presynaptic accumulator v_k ,

$$\tau_v \frac{dv_k}{dt} = -v_k(t) + \phi x_k(t), \quad (4)$$

and a postsynaptic accumulator u_k that averages over nearby synapses in a weighted and distance-dependent manner,

$$\tau_u \frac{du_k}{dt} = -u_k(t) + \sum_{l=1}^N s_{kl} w_l(t) x_l(t). \quad (5)$$

The multiplicative factor ϕ is an MMP9 efficiency constant that determines how efficiently MMP9 converts proBDNF into BDNF per unit of time and the proximity variables s_{kl} between synapses k and l on the same dendrite are computed as $s_{kl} = e^{-\frac{d_{kl}^2}{2\sigma^2}}$, where σ determines the spatial postsynaptic calcium spread constant. The equation governing the weight development of w_k (Equation 6) is a Hebbian equation that directly combines the pre- and postsynaptic accumulator with an additional offset constant ρ ,

$$\tau_w \dot{w}_k = u_k(t)(v_k(t) + \rho), \quad (6)$$

with $\rho = \frac{2\eta-1}{2(1-\eta)}$ and $\tau_w = \tau_W \frac{1}{2(1-\eta)}$. Here, η is the constitutive ratio of BDNF to proBDNF and $\tau_W = 3000$ ms. This model is minimal in the sense that it cannot be reduced further without losing either the dependence on correlation through the link to the BTDP rule, or the dependence on distance.

To model structural plasticity, we implemented a structural plasticity rule inspired by ref. (Holtmaat and Svoboda, 2009) where each synapse whose efficacy falls below a fixed threshold w_{thr} is pruned from the dendrite.

Simulations and parameters

For all simulations in this study, we distributed nine postsynaptic somata at regular distances in a grid formation on a 2-dimensional 185×185 pixel area, representing a cortical sheet (where 1 pixel = 1 micron, [Figure 4](#)). This yields a density of around 300 neurons per mm^2 (translating to around 5,000 per mm^3 , where for 25 neurons in [Figure 3-Figure Supplement 2](#) this would be around 750 neurons per mm^2 or 20,000 per mm^3). The explored densities are a bit lower than compared to neuron densities observed in adulthood ([Keller et al., 2018](#)).

In the same grid, we randomly distributed 1,500 potential synapses, yielding an initial density of 0.044 potential synapses per μm^2 ([Figure 3e](#)). At the end of the simulation time, around 1,000 potential synapses remain, showing that the density of potential synapses is sufficient and does not significantly affect the final density of connected synapses. Thus, the rapid slowing down of growth in our model is not due to a depletion of potential synaptic partners. The resulting density of stably connected synapses is approximately 0.015 synapses per μm^2 (around 60 synapses stabilized per dendritic tree, [Figure 3b](#)). This density compares well to experimental findings, where, especially during early development, synaptic densities are described to be within a range similar to the one observed in our model ([Leighton et al., 2024](#); [Ultanir et al., 2007](#); [Glynn et al., 2011](#); [Yang et al., 2014](#); [Koshimizu et al., 2009](#); [Tyler and Pozzo-Miller, 2001](#)). The potential synapses were divided into five groups of equal size (300 synapses per group), with each group receiving Poisson input with rate r_{in} . Therefore, all synapses in the same group are perfectly correlated, while synapses in different groups are uncorrelated.

Parameters, as the somata density, activity group correlations or dynamic potential synapses, were only changed for simulations in which we explicitly explored their effects on the models behavior (see [Figure 3-Figure Supplement 2](#), [Figure 3-Figure Supplement 3](#), [Figure 5](#), [Figure 5-Figure Supplement 2](#), and sections below).

Within-group and across-group activity correlations. For the decreased within-group correlations, we generated parent spike trains for each individual group with the firing rate $r_{in} = r_{total} * P_{in}$ (e.g., $P_{in} = 100\%$; 60% ; 20% , [Figure 5](#)). All the synapses of the same group share the same parent spike train and the remaining spikes for each synapse are uniquely generated with the firing rate $r_{rest} = r_{total} * (1 - P_{in})$ (e.g., $(1 - P_{in}) = 0\%$; 40% ; 80%), resulting in the desired firing rate r_{cross} (see [Table 1](#)). For the increase in across-group correlations, we generated one master spike train with the firing rate $r_{cross} = r_{total} * P_{cross}$ for all the synapses of all groups (e.g., $P_{cross} = 5\%$; 10% ; 20% , [Figure 5-Figure Supplement 2](#)). This master spike train is shared across all groups and then filled up according to the within-group correlation (if not specified differently $P_{in} = 1 - P_{cross}$ to maintain the rate r_{total}). In all the cases, also in those where across-group correlations are combined with within-group correlations, the remaining spikes for each synapse are generated with a firing rate $r_{rest} = r_{total} * (1 - P_{in} - P_{cross})$, to obtain an overall desired firing rate of r_{total} is obtained.

Dynamic potential synapses

To implement dynamic potential synapses we introduced life-times for each synapse, randomly sampled from a log-normal distribution with median 1.8 h (for when they move frequently), 4.5 h or 18 h (for when they move rarely) and variance equal to 1 ([Figure 3-Figure Supplement 3b](#)). The lifetime of a synapse decreases only when the synapse is not connected to any of the dendrites (i.e., is a potential synapse). When the lifetime of a synapse expires, the synapse moves to a new location with a new lifetime sampled from the same log-normal distribution. This enables synapses to move multiple times throughout a simulation. The exact locations and distances to which each synapse can move are determined by a binary matrix (dimensions: $pixel_{distance} \times pixel_{distance}$) representing a ring (annulus) with the inner radius $\frac{r}{2}$ and outer radius $\frac{r}{2}$, where the synapse location is at the center of the matrix. All the locations of the matrix within the ring

boundaries (between inner radius and outer radius) are potential locations to which the synapse can move. The synapse then moves randomly to one of the possible locations where no other synapse or dendrite is located. For the movement distances, we chose the ring dimensions 3×3 , 25×25 and 101×101 , yielding the moving distances (radii) of 1 pixel per movement, 12 pixels per movement and 50 pixels per movement ($r = (d-1)/2$). These pixel distances represent small movements, as much as a dendrite can grow in one step (1 micron), and larger movements which are far enough so that the synapse will not attract the same branches again (12 microns) or far enough so that it might attract a completely different dendrite (50 microns, *Figure 3-Figure Supplement 3a*).

Comparison with wiring optimization MST models

To evaluate the wire minimization properties of our model morphologies ($n=288$), we examined whether the number of connected synapses (N), total length (L), and surface area of the spanning field (S) conformed to the scaling law $L \approx \pi^{-1/2} \cdot S^{1/2} \cdot N^{1/2}$ (Cuntz et al., 2012 [↗](#)). Furthermore, to validate that our model dendritic morphologies scale according to optimal wiring principles, we created simplified models of dendritic trees using the MST algorithm with a balancing factor (bf). This balancing factor adjusts between minimizing the total dendritic length and minimizing the tree distance between synapses and the soma ($\text{Cost} = L + bf \cdot P \cdot D$) (MST_tree; best $bf = 0.925$) (Cuntz et al., 2010 [↗](#)); TREES Toolbox <http://www.treestoolbox.org> [↗](#)). Initially, we generated MSTs to connect the same distributed synapses as our models. We performed MST simulations that vary the balancing factor between $bf = 0$ and $bf = 1$ in steps of 0.025 while calculating the morphometric agreement by computing the error (Euclidean distance) between the morphologies of our models and those generated by the MST models. The morphometrics used were total length, number of terminals, and surface area occupied by the synthetic morphologies.

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Data availability

All code can be accessed at: <https://github.com/comp-neural-circuits/dendritic-growth> [↗](#)

Competing interests

No competing interests declared.

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Reviewer #2 (Public review):

This work combines a model of two-dimensional dendritic growth with attraction and stabilisation by synaptic activity. The authors find that constraining growth models with competition for synaptic inputs produces artificial dendrites that match some key features of real neurons both over development and in terms of final structure. In particular, incorporating distance-dependent competition between synapses of the same dendrite naturally produces distinct phases of dendritic growth (overshoot, pruning, and stabilisation) that are observed biologically and leads to local synaptic organisation with functional relevance. The approach is elegant and well-explained but makes some significant modelling assumptions that might impact the biological relevance of the results.

The main strength of the work is the general concept of combining morphological models of growth with synaptic plasticity and stabilisation. This is an interesting way to bridge two distinct areas of neuroscience in a manner that leads to findings that could be significant for both. The modelling of both dendritic growth and distance-dependent synaptic competition is carefully done, constrained by reasonable biological mechanisms, and well-described in the text. The paper also links its findings, for example in terms of phases of dendritic growth or final morphological structure, to known data well.

The authors have managed to address my previous comments on the paper well by considering axonal dynamics, spatial correlations, and the effects of changing ratios of BDNF-proBDNF. The modelling has now been validated over a wider range of confounding factors and looks to be a solid basis for future work in this direction.

<https://doi.org/10.7554/eLife.87527.2.sa2>

Reviewer #3 (Public review):

The authors propose a mechanistic model of how the interplay between activity-independent growth and an activity-dependent synaptic strengthening/weakening model influences the dendrite shape, complexity, and distribution of synapses. The authors focus on a model for stellate cells with multiple dendrites emerging from a soma. The activity-independent component is provided by a random pool of presynaptic sites representing potential synapses and releasing a diffusible signal promoting dendritic growth. Then, a spontaneous activity pattern with some correlation structure is imposed at those presynaptic sites. The strength of these synapses follows a learning rule previously proposed by the lab: synapses strengthen when there is correlated firing across multiple sites, and synapses weaken if there is uncorrelated firing with the relative strength of these processes controlled by available levels of BDNF/proBDNF. Once a synapse is weakened below a threshold, the dendrite branch at that site retracts and loses its sensitivity to the growth signal.

This revised version of the manuscripts contains clarifications and additional experiments that better reflect the robustness of the model. I continue to maintain my favorable review. I am glad the research persevered the long delays with changing trainees.

<https://doi.org/10.7554/eLife.87527.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public Review):

The authors introduce a computational model that simulates the dendrites of developing neurons in a 2D plane, subject to constraints inspired by known biological mechanisms such as diffusing trophic factors, trafficked resources, and an activity-dependent pruning rule. The resulting arbors are analyzed in terms of their structure, dynamics, and responses to certain manipulations. The authors conclude that 1) their model recapitulates a stereotyped timecourse of neuronal development: outgrowth, overshoot, and pruning 2) Neurons achieve near-optimal wiring lengths, and Such models can be useful to test proposed biological mechanisms- for example, to ask whether a given set of growth rules can explain a given observed phenomenon - as developmental

neuroscientists are working to understand the factors that give rise to the intricate structures and functions of the many cell types of our nervous system.

Overall, my reaction to this work is that this is just one instantiation of many models that the author could have built, given their stated goals. Would other models behave similarly? This question is not well explored, and as a result, claims about interpreting these models and using them to make experimental predictions should be taken warily. I give more detailed and specific comments below.

We thank the reviewer for the summary of the work. But the criticism “that this is one instantiation of many models [we] could have built” is unfair as it can apply to any model. We chose one of the most minimalistic models which implements known biological mechanisms including activity-independent and -dependent phases of dendritic growth, and constrained parameters based on experimental data. We compare the proposed model to other alternatives in the Discussion section. In the revised manuscript, we additionally investigate the sensitivity of model output to variations of specific parameters, as explained below.

Point 1.1. Line 109. After reading the rest of the manuscript, I worry about the conclusion voiced here, which implies that the model will extrapolate well to manipulations of all the model components. How were the values of model parameters selected? The text implies that these were selected to be biologically plausible, but many seem far off. The density of potential synapses, for example, seems very low in the simulations compared to the density of axons/boutons in the cortex; what constitutes a potential synapse? The perfect correlations between synapses in the activity groups is flawed, even for synapses belonging to the same presynaptic cell. The density of postsynaptic cells is also orders of magnitude off, etc. Ideally, every claim made about the model's output should be supported by a parameter sensitivity study. The authors performed few explorations of parameter sensitivity and many of the choices made seem ad hoc.

We have performed detailed sensitivity analysis on the model parameters mentioned by the reviewer, including (I) the density of postsynaptic cells (somatas), (II) the density of potential synapses, and (III) the level of correlations between synapses.

(I) While the density of postsynaptic cells in our baseline model seems a bit low, at least when compared to densities observed in adulthood (Keller et al., 2018), we explored how altering this value affects the model dynamics. We found that the postsynaptic cell density does not affect the timing of dendritic outgrowth, overshoot and synaptic pruning. It only changes the final size of the dendritic arbor and the resulting number of connected synapses. This analysis is now included in Supplementary Figure 3-2.

(II) The density of potential synapses and the density of connected synapses that we used in the manuscript are already in the range of densities that can be found in the literature (Leighton et al., 2024; Ultanir et al., 2007; Glynn et al., 2011; Yang et al., 2014), some of which we already cited in the original submission.

A potential concern might be that the rapid slowing down of growth in the model could be due to a depletion of potential synapses. To illustrate that this is not the case, we showed that the number of available potential synapses over the time course of the simulations remains high (Figure 3, new panel e). Therefore, the initial density of potential synapses is sufficient and does not affect the final density of connected synapses.

To further illustrate the robustness of our model dynamics to longer simulation times, we added a new supplementary figure (Supplementary Figure 3-1).

These new figure additions (Figure 3e, Supplementary Figure 3-1, and Supplementary Figure 3-2) and their implications for the model dynamics are discussed in the Results section of the

revised paper:

p.9 line 198, “After the initial overshoot and pruning, dendritic branches in the model stay stable, with mainly small subbranches continuing to be refined (Figure 3-Figure Supplement 1). This stability in the model is achieved despite the number of potential synaptic partners remaining high (Figure 3e), indicating a balance between activity-independent and activity-dependent mechanisms. The dendritic growth and synaptic refinement dynamics are independent of the postsynaptic somata densities used in our simulations (Figure 3-Figure Supplement 2). Only the final arbor size and the number of connected synapses decrease with an increase in the density of the somata, while the timing of synaptic growth, overshoot and pruning remains the same (Figure 3-Figure Supplement 2).”

We also added more details to the description of our model in the Methods section:

p.24 line 615, “For all simulations in this study, we distributed nine postsynaptic somata at regular distances in a grid formation on a 2-dimensional 185×185 pixel area, representing a cortical sheet (where 1 pixel = 1 micron, Figure 4). This yields a density of around 300 neurons per mm^2 (translating to around 5,000 per mm^3 , where for 25 neurons in Figure 3-Figure Supplement 2 this would be around 750 neurons per mm^2 or 20,000 per mm^3). The explored densities are a bit lower than compared to neuron densities observed in adulthood (Keller et al., 2018). In the same grid, we randomly distributed 1,500 potential synapses, yielding an initial density of 0.044 potential synapses per μm^2 (Figure 3e). At the end of the simulation time, around 1,000 potential synapses remain, showing that the density of potential synapses is sufficient and does not significantly affect the final density of connected synapses. Thus, the rapid slowing down of growth in our model is not due to a depletion of potential synaptic partners. The resulting density of stably connected synapses is approximately 0.015 synapses per μm^2 (around 60 synapses stabilized per dendritic tree, Figure 3b). This density compares well to experimental findings, where, especially during early development, synaptic densities are described to be within a range similar to the one observed in our model (Leighton et al., 2024; Ultanir et al., 2007; Glynn et al., 2011; Yang et al., 2014; Koshimizu et al., 2009; Tyler and Pozzo-Miller, 2001).”

(III) Lastly, we investigated how the correlation between synapses of the same activity group might affect our conclusions. As correlations in our model mainly arise from patterns of spontaneous activity which are abundant in early postnatal development (retinal waves (Ackman et al., 2012) or endogenous activity in the form of highly synchronized events involving a large fraction of the cells (Siegel et al., 2012), we explored varying the correlations within each activity group, across activity groups and combinations of both. While this analysis supported our previously described intuition on how competition between synaptic activities should drive activity-dependent refinement, recently a study found direct evidence for such subcellular refinement of synaptic inputs specifically dependent on spontaneous activity between retinal ganglion cell axons and retinal waves in the superior colliculus (Matsumoto et al., 2024). The new analysis confirmed our earlier results that the competition between activity groups leads to activity-dependent refinement and yielded further insight into how the studied activity correlations can affect the competition. Those results are presented in a completely new figure (new Figure 5, supported by the Supplementary Figure 5-1 and 5-2) and discussed in the Results section:

p.11 line 249, “Group activity correlations shape synaptic overshoot and selectivity competition across synaptic groups.

Since correlations between synapses emerge from correlated patterns of spontaneous activity abundant during postnatal development (Ackman et al., 2012; Siegel et al., 2012), we explored a wide range of within-group correlations in our model (Figure 5a). Although a change in correlations within the group has only a minor effect on the resulting dendritic lengths (Figure 5b) and overall dynamics, it can change the density of connected synapses and thus

also affect the number of connected synapses to which each dendrite converges throughout the simulations (Figure 5c,e). This is due to the change in specific selectivity of each dendrite which is a result of the change in within-group correlations (Figure 5d). While it is easier for perfectly correlated activity groups to coexist within one dendrite (Figure 5-Figure Supplement 1a, 100%), decreasing within-group correlations increases the competition between groups, producing dendrites that are selective for one specific activity group (60%, Figure 5d, Figure 5-Figure Supplement 1a). This selectivity for a particular activity group is maximized at intermediate (approximately 60%) within-group correlations, while the contribution of the second most abundant group generally remains just above random chance levels (Figure 5-Figure Supplement 1a). Further reducing within-group correlations (20%, Figure 5a) causes dendrites to lose their selectivity for specific activity groups due to the increased noise in the activity patterns (20%, Figure 5a). Overall, reducing within-group correlations increases synapse pruning (Figure 5f, bottom), also found experimentally (Matsumoto et al., 2024) as dendrites require an extended period to fine-tune connections aligned with their selectivity biases. This phenomenon accounts for the observed reduction in both the density and number of synapses connected to each dendrite.

In addition to the within-group correlations, developmental spontaneous activity patterns can also change correlations between groups as for example retinal waves propagated in different domains (Feller et al., 1997) (Figure 5-Figure Supplement 2). An increase in between-group correlations in our model intuitively decreases competition between the groups since fully correlated global events synchronize the activity of all groups (Figure 5-Figure Supplement 2). The reduction in competition reduces pruning in the model, which can be recovered by combining cross-group correlations with decreased within-group correlations (Figure 5-Figure Supplement 2). Our simulations show that altering the correlations within activity groups increases competition (by lowering the within-group correlations) or decreases competition (by raising the across-group correlations). Hence, in our model, competition between activity groups due to non-trivially structured correlations is necessary to generate realistic dynamics between activity-independent growth and activity-dependent refinement or pruning.

In sum, our simulations demonstrate that our model can operate under various correlations in the spike trains. We find that the level of competition between synaptic groups is crucial for the activity-dependent mechanisms to either potentiate or depress synapses and is fully consistent with recent experimental evidence showing that the correlation between spontaneous activity in retinal ganglion cells axons and retinal waves in the superior colliculus governs branch addition vs. elimination (Matsumoto et al., 2024)."

Precise details on the implementation of the changed activity correlations were added to the Methods section:

p. 25 line 638, "Within-group and across-group activity correlations. For the decreased within-group correlations, we generated parent spike trains for each individual group with the firing rate $r_{in} = r_{total} * P_{in}$ (e.g., $P_{in} = 100\%$; 60% ; 20% , Figure 5). All the synapses of the same group share the same parent spike train and the remaining spikes for each synapse are uniquely generated with the firing rate $r_{rest} = r_{total} * (1 - P_{in})$ (e.g., $(1 - P_{in}) = 0\%$; 40% ; 80%), resulting in the desired firing rate r_{total} (see Table 1). For the increase in across-group correlations, we generated one master spike train with the firing rate $r_{cross} = r_{total} * P_{cross}$ for all the synapses of all groups (e.g., $P_{cross} = 5\%$; 10% ; 20% , Figure 5-Figure Supplement 2). This master spike train is shared across all groups and then filled up according to the within-group correlation (if not specified differently $P_{in} = 1 - P_{cross}$ to maintain the rate r_{total}). In all the cases, also in those where the change in across-group correlations is combined with the change in within-group correlations, the remaining spikes for each synapse are generated with a firing rate $r_{rest} = r_{total} * (1 - P_{in} - P_{cross})$ to obtain an overall desired firing rate of r_{total} ."

Point 1.2. Many potentially important phenomena seem to be excluded. I realize that no model can be complete, but the choice of which phenomena to include or exclude from this model could bias studies that make use of it and is worth serious discussion. The development of axons is concurrent with dendrite outgrowth, is highly dynamic, and perhaps better understood mechanistically. In this model, the inputs are essentially static. Growing dendrites acquire and lose growth cones that are associated with rapid extension, but these do not seem to be modeled. Postsynaptic firing does not appear to be modeled, which may be critical to activity-dependent plasticity. For example, changes in firing are a potential explanation for the global changes in dendritic pruning that occur following the outgrowth phase.

Thanks to the reviewer for bringing up these important considerations. We do indeed write in the Introduction (e.g. lines 36-76) which phenomena we include in the model and why. The Discussion also compares our model to others (lines 433-490), pointing out that most models either focus on activity-independent or activity-dependent phases. We include both, combining the influence of both molecular gradients and growth factors as well as activity-dependent connectivity refinements instructed by spontaneous activity. We consider our model a tractable, minimalist mechanistic model which includes both activity-independent and activity-dependent aspects.

Regarding postsynaptic firing, this is indeed super relevant and an important point to consider. In one of our recent publications (Kirchner and Gjorgjieva, 2021), we studied only an activity-dependent model for the organization of synaptic inputs on non-growing dendrites which have a fixed length. There, we considered the effect of postsynaptic firing (via a back-propagating action potential) and demonstrated that it plays an important role in establishing a global organization of synapses on the entire dendritic tree of the neuron. For example, we showed that it could lead to the emergence of retinotopic maps on the dendritic tree which have been found experimentally (Iacaruso et al., 2017). Since we use the same activity-dependent plasticity model in this paper, we expect that the somatic firing will have the same effect on establishing synaptic distributions on the entire dendritic tree. This is now also discussed in the Discussion section of the revised manuscript:

p. 21 line 491, “Although we did not explicitly model postsynaptic firing, our previous work with static dendrites has shown that it can play an important role in establishing a global organization of synapses on the entire dendritic tree of the neuron (Kirchner and Gjorgjieva, 2021). For example, we showed that it could lead to the emergence of retinotopic maps on the dendritic tree which have been found experimentally (Iacaruso et al., 2017). Since we use the same activity-dependent plasticity model in this paper, we expect that the somatic firing will have the same effect on establishing synaptic distributions on the entire dendritic tree.”

Including the concurrent development of axons in the model is indeed very interesting. In fact, a recent tour-de-force techniques paper found similar to what we assume. Hebbian activity-dependent dynamics of axonal branches of retinal ganglion cells experiencing spontaneous activity in relation to retinal waves in the superior colliculus (Matsumoto et al., 2024). New branches tend to be added at the locations where spontaneous activity of individual branches is more correlated with retinal waves, whereas asynchronous activity is associated with branch elimination. We suspect the same Hebbian activity-dependent dynamics to apply also to dendritic growth.

To address simultaneous dynamic axons to our growing dendrites, in the revised version of the manuscript, we included a simplified form of axonal dynamics by allowing changes in the lifetime and location of potential synapses, which come from axons of presynaptic partners. We explored different median lifetimes of synapses in combination with several distances with which a synapse can move in the simulated space (new Supplementary Figure 3-3). Our

results show that dynamically moving synapses only affect the dynamics and stability of our model when the rate of moving synapses combined with the distance of moving synapses is faster than the dendritic growth. In scenarios in which synapses can move across large distances, dendrites get further destabilized due to synapses transferring from one dendrite to another, perturbing the attractor fields of the potential synapses even in late phases of the simulations. Besides such non-biological scenarios, dynamically moving synapses do not affect the model dynamics too much. Thus, they mostly add additional noise and variability to the growth and pruning without changing the timing and amplitude of the dynamics. These results are discussed in the results section of the revised manuscript:

p.9 line 207, “The development of axons is concurrent with dendritic growth and highly dynamic Matsumoto et al. (2024). To address the impact of simultaneously growing axons, we implemented a simple form of axonal dynamics by allowing changes in the lifetime and location of potential synapses, originating from the axons of presynaptic partners (Figure 3–Figure Supplement 3). When potential synapses can move rapidly (median lifetime of 1.8 hours), the model dynamics are perturbed quite substantially, making it difficult for the dendrites to stabilize completely (Figure 3–Figure Supplement 3c). However, slowly moving potential synapses (median lifetime of 18 hours) still yield comparable results (Figure 3–Figure Supplement 3). The distance of movement significantly influenced results only when potential synaptic lifetimes were short. For extended lifetimes, the moving distance had a minor impact on the dynamics, predominantly affecting the time required for dendrites to stabilize. This was the result of synapses being able to transfer from one dendrite to another, potentially forming new long-lasting connections even at advanced stages of synaptic refinement. In sum, our results show that potential axonal dynamics only affect the stability of our model when these dynamics are much faster than dendritic growth.”

Precise details on the implementation of the dynamically moving synapses and their synaptic lifetimes are now in the Methods section:

p. 25 line 650, “Dynamically moving synapses. For the moving synapses we introduced lifetimes for each synapse, randomly sampled from a log-normal distribution with median 1.8h (for when they move frequently), 4.5h or 18h (for when they move rarely) and variance equal to 1 (Figure 3–Figure Supplement 3b). The lifetime of a synapse decreases only when the synapse is not connected to any of the dendrites (i.e., is a potential synapse). When the lifetime of a synapse expires, the synapse moves to a new location with a new lifetime sampled from the same log-normal distribution. This enables synapses to move multiple times throughout a simulation. The exact locations and distances to which each synapse can move are determined by a binary matrix (dimensions: $\text{pixeldistance} \times \text{pixeldistance}$) representing a ring (annulus) with the inner radius $d/4$ and outer radius $d/2$, where the synapse location is at the center of the matrix. All the locations of the matrix within the ring boundaries (between the inner radius and outer radius) are potential locations to which the synapse can move. The synapse then moves randomly to one of the possible locations where no other synapse or dendrite is located. For the movement distances, we chose the ring dimensions 3×3 , 25×25 and 101×101 , yielding the moving distances (radii) of 1 pixel per movement, 12 pixels per movement and 50 pixels per movement ($r = (d-1)/2$). These pixel distances represent small movements, as much as a dendrite can grow in one step (1 micron), and larger movements which are far enough so that the synapse will not attract the same branches again (12 microns) or far enough so that it might attract a completely different dendrite (50 microns, Figure 3–Figure Supplement 3a).”

Point 1.3. Line 167. There are many ways to include activity -independent and -dependent components into a model and not every such model shows stability. A key feature seems to be that larger arbors result in reduced growth and/or increased retraction, but this could be achieved in many ways (whether activity dependent or not). It's not clear that this result is due to the combination of activity-

dependent and independent components in the model, or conceptually why that should be the case.

We never argued for model uniqueness. There are always going to be many different models (at different spatial and temporal scales, at different levels of abstraction). We can never study all of them and like any modeling study in systems neuroscience we have chosen one model approach and investigated this approach. We do compare the current model to others in the Discussion. If the reviewers have a specific implementation that we should compare our model to as an alternative, we could try, but not if this means doing a completely separate project.

Point 1.4. Line 183. The explanation of overshoot in terms of the different timescales of synaptic additions versus activity-dependent retractions was not something I had previously encountered and is an interesting proposal. Have these timescales been measured experimentally? To what extent is this a result of fine-tuning of simulation parameters?

We found that varying the amount of BDNF controls the timescale of the activity-dependent plasticity (see our Figure 6c). Hence, changing the balance between synaptic additions vs. retractions is already explored in Figure 6e and f. Here we show that the overshoot and retraction does not have to be fine-tuned but may be abolished if there is too much activity-dependent plasticity.

Regarding the relative timescales of synaptic additions vs. retractions: since the first is mainly due to activity-independent factors, and the second due to activity-dependent plasticity, the questions is really about the timescales of the latter two. As we write in the Introduction (lines 61-63), manipulating activity-dependent synaptic transmission has been found to not affect morphology but rather the density and specificity of synaptic connections (Ultanir et al. 2007), supporting the sequential model we have (although we do not impose the sequence, as both activity-independent and activity-dependent mechanisms are always “on”; but note that activity-dependent plasticity can only operate on synapses that have already formed).

The described results are robust to parameter variations (performed on the postsynaptic density, potential synapse density, and within- and across-group correlations) as described in the reply to reviewer #1 point 1.1.

Point 1.5. Line 203. This result seems at odds with results that show only a very weak bias in the tuning distribution of inputs to strongly tuned cortical neurons (e.g. work by Arthur Konnerth's group). This discrepancy should be discussed.

First, we note that the correlated activity experienced by our modeled synapses (and resulting synaptic organization) does not necessarily correspond to visual orientation, or any stimulus feature, for that matter, but is rather a property of correlated spontaneous activity.

Nonetheless, there is some variability in what the experimental data show. Many studies have shown that synapses on dendrites are organized into functional synaptic clusters: across brain regions, developmental ages and diverse species from rodent to primate (Kleindienst et al., 2011; Takahashi et al., 2012; Winnubst et al., 2015; Gökçe et al., 2016; Wilson et al., 2016; Iacaruso et al., 2017; Scholl et al., 2017; Niculescu et al., 2018; Kerlin et al., 2019; Ju et al., 2020, Hedrick et al., 2022, Hedrick et al., 2024). Interestingly, some *in vivo* studies have reported lack of fine-scale synaptic organization (Varga et al., 2011; X. Chen et al., 2011; T-W. Chen et al., 2013; Jia et al., 2010; Jia et al., 2014), while others reported clustering for different stimulus features in different species. For example, dendritic branches in the ferret visual cortex exhibit local clustering of orientation selectivity but do not exhibit global organization of inputs according to spatial location and receptive field properties (Wilson et al. 2016;

Scholl et al., 2017). In contrast, synaptic inputs in mouse visual cortex do not cluster locally by orientation, but only by receptive field overlap, and exhibit a global retinotopic organization along the proximal-distal axis (Iacaruso et al., 2017). We proposed a theoretical framework to reconcile these data: combining activity-dependent plasticity similar to the BDNF-proBDNF model that we used in the current work, and a receptive field model for the different species (Kirchner and Gjorgjieva, 2021). This is now also discussed in the Discussion section of the revised manuscript:

p. 20 line 471, “The correlated activity experienced by our modeled synapses (and resulting synaptic organization) does not necessarily correspond to visual orientation, or any stimulus feature, for that matter, but is rather a property of spontaneous activity. Nonetheless, there is some variability in what the experimental data show. Many have shown that synapses on dendrites are organized into functional synaptic clusters: across brain regions, developmental ages and diverse species from rodent to primate (Kleindienst et al., 2011; Winnubst et al., 2015; Iacaruso et al., 2017; Scholl et al., 2017; Niculescu et al., 2018; Takahashi et al., 2012; Gökçe et al., 2016; Wilson et al., 2016; Kerlin et al., 2019; Ju et al., 2020; Hedrick et al., 2022, 2024). Other studies have reported lack of fine-scale synaptic organization (Chen et al., 2013; Varga et al., 2011; Chen et al., 2011; Jia et al., 2010, 2014). Interestingly, some of these discrepancies might be explained by different species showing clustering with respect to different stimulus features (orientation or receptive field overlap) (Scholl et al., 2017; Wilson et al., 2016; Iacaruso et al., 2017). Our prior work proposed a theoretical framework to reconcile these data: combining activity-dependent plasticity as we used in the current work, and a receptive field model for the different species (Kirchner and Gjorgjieva, 2021).”

Point 1.6. Line 268. How does the large variability in the size of the simulated arbors relate to the relatively consistent size of arbors of cortical cells of a given cell type? This variability suggests to me that these simulations could be sensitive to small changes in parameters (e.g. to the density or layout of presynapses).

We again thank the reviewer for the detailed explanation and feedback on parameters that should be tested in more detail. We have explored several of the suggested model parameters and believe that we have managed to explain and illustrate their effects on the model's dynamics clearly. The precise changes are explained in the reply to point 1.1 and are now available in the revised version of the manuscript.

Point 1.7. The modeling of dendrites as two-dimensional will likely limit the usefulness of this model. Many phenomena- such as diffusion, random walks, topological properties, etc - fundamentally differ between two and three dimensions.

Indeed, there are many differences between two and three dimensions. We have ongoing work that extends the current model to 3D but is beyond the scope of the current paper. In systems neuroscience, people have found very interesting results making such simplified geometric assumptions about networks, for instance the one-dimensional ring model has been used to uncover fundamental insights about computations even though highly simplified and abstracted. We are convinced that our model, especially with the new sensitivity analysis, makes interesting and novel contributions and predictions.

Point 1.8. The description of wiring lengths as 'approximately optimal' in this text is problematic. The plotted data show that the wiring lengths are several deviations away from optimal, and the random model is not a valid instantiation of the 2D non-overlapping constraints the authors imposed. A more appropriate null should be considered.

We appreciate the reviewer's feedback regarding the use of the term "approximately optimal" in describing wiring lengths. We acknowledge that our initial terminology was imprecise and could be misleading. We had previously referred to the minimal wiring length as the optimal wiring length, which does not fully capture the nuances of neuronal wiring optimization. As noted in prior literature, such as the work by Hermann Cuntz (Cuntz et al., 2010 & 2012), neurons can optimize their wiring beyond simply minimizing dendritic length.

To address this issue, to better capture the balance between wiring minimization and functional constraints, such as conduction delays, we have developed a new modeling approach based on minimum spanning trees with a balancing factor (Cuntz et al., 2010 & 2012). This factor modulates the trade-off between minimizing wiring length and accounting for conduction delays from synapses to the soma. Specifically, the model assumes a balance between minimizing the total dendritic length and minimizing the tree distance between synapses and the site of input integration, typically the soma. This balance is illustrated in Figure 8 (Figure 7 in the original manuscript), where we demonstrate that the deviation from the theoretical minimum length arises because direct paths to synapses often require longer dendrites in our models.

Together with the new result, which we added as the new panels f, g and h to Figure 8 (originally Figure 7), we also adjusted panel a of Figure 8, to now illustrate the difference between random wiring, minimal wiring and minimal conductance delay. The updated Figure 8 and its new findings are discussed in the results section of the revised manuscript:

p.17 line 387, "This deviation is expected given that real dendrites need to balance their growth processes between minimizing wire while reducing conduction delays. The interplay between these two factors emerges from the need to reduce conduction delays, which requires a direct path length from a given synapse to the soma, consequently increasing the total length of the dendritic cable. (Cuntz et al., 2010, 2012; Ferreira Castro et al., 2020).

To investigate this further, we compared the scaling relations of the final morphologies of our models with other synthetic dendritic morphologies generated using a previously described minimum spanning tree (MST) based model. The MST model balances the minimization of total dendritic length and the minimization of conduction delays between synapses and the soma. This balance results in deviations from the theoretical minimum length because direct paths to synapses often require longer dendrites (Cuntz et al., 2008, 2010). The balance in the model is modulated by a balancing factor (bf). If bf is zero, dendritic trees minimize the cable only, and if bf is one, they will try to minimize the conduction delays as much as possible. It is important to note that the MST model does not simulate the developmental process of dendritic growth; it is a phenomenological model designed to generate static morphologies that resemble real cells.

To facilitate the comparison of total lengths between our simulated and MST morphologies, we generated MST models under the same initial conditions (synaptic spatial distribution) as our models and simulated them to match several morphometrics (total length, number of terminals, and surface area) of our grown morphologies. This allowed us to create a corresponding MST tree for each of our synthetic trees. Consequently, we could evaluate whether the branching structures of our models were accurately predicted by minimum spanning trees based on optimal wiring constraints. We found that the best match occurred with a trade-off parameter $bf = 0.9250$ (Figure 8f). Using the morphologies generated by the MST model with the specified trade-off parameter (bf), we showed that the square root of the synapse count and the total length (L) in both our model generated trees and the MST trees exhibit a linear scaling relationship (Figure 8g; $R^2 = 0.65$). The same linear relationship can be observed for the square root of the surface area and the total length L of our model trees and the MST trees (Figure 8h; $R^2 = 0.73$). Overall, these results indicate that our model generate trees are wellfitted by the MST model and follow wire optimization constraints.

We acknowledge that the value of the balancing factor bf in our model is higher than the range of balancing factors that is typically observed in the biological dendritic counterparts, which generally ranges between 0.2 and 0.4 (Cuntz et al., 2012; Ferreira Castro et al., 2020; Baltruschat et al., 2020). However, it is still remarkable that our model, which does not explicitly address these two conservation laws, achieves approximately optimal wiring. Why do we observe such a high bf value? We reason that two factors may contribute to this. First, in our models, local branches grow directly to the nearest potential synapse, potentially taking longer routes instead of optimally branching to minimize wiring length (Wen and Chklovskii, 2008). Second, the growth process in our models does not explicitly address the tortuosity of the branches, which can increase the total length of the branches used to connect synapses. In the future, it will be interesting to add constraints that take these factors into account. Taken together, combining activity-independent and -dependent dendrite growth produces morphologies that approximate optimal wiring.”

Further details on the fitted MST model and the corresponding analysis were added to the methods section:

p.26 line 669, “Comparison with wiring optimization MST models. To evaluate the wire minimization properties of our model morphologies ($n=288$), we examined whether the number of connected synapses (N), total length (L), and surface area of the spanning field (S) conformed to the scaling law $L \approx \pi^{-1/2} \cdot S^{1/2} \cdot N^{1/2}$ (Cuntz et al., 2012). Furthermore, to validate that our model dendritic morphologies scale according to optimal wiring principles, we created simplified models of dendritic trees using the MST algorithm with a balancing factor (bf). This balancing factor adjusts between minimizing the total dendritic length and minimizing the tree distance between synapses and the soma ($\text{Cost} = L + bf \cdot P \cdot L$) (MST_tree; best $bf = 0.925$) (Cuntz et al., 2010); TREES Toolbox <http://www.treestoolbox.org>).

Initially, we generated MSTs to connect the same distributed synapses as our models. We performed MST simulations that vary the balancing factor between $bf = 0$ and $bf = 1$ in steps of 0.025 while calculating the morphometric agreement by computing the error (Euclidean distance) between the morphologies of our models and those generated by the MST models. The morphometrics used were total length, number of terminals, and surface area occupied by the synthetic morphologies.”

Point 1.9. It's not clear to me what the authors are trying to convey by repeatedly labeling this model as 'mechanistic'. The mechanisms implemented in the model are inspired by biological phenomena, but the implementations have little resemblance to the underlying biophysical mechanisms. Overall my impression is that this is a phenomenological model intended to show under what conditions particular patterns are possible. Line 363, describing another model as computational but not mechanistic, was especially unclear to me in this context.

What we mean by mechanistic is that we implement equations that model specific mechanisms i.e. we have a set of equations that implement the activity-independent attraction to potential synapses (with parameters such as the density of synapses, their spatial influence, etc) and the activitydependent refinement of synapses (with parameters such as the ratio of BDNF and proBDNF to induce potentiation vs depression, the activity-dependent conversion of one factor to the other, etc). This is a bottom-up approach where we combine multiple elements together to get to neuronal growth and synaptic organization. This approach is in stark contrast to the so-called top-down or normative approaches where the method would involve defining an objective function (e.g. minimal dendritic length) which depends on a set of parameters and then applying a gradient descent or other mathematical optimization technique to get at the parameters that optimize the objective function. This latter approach we would not call mechanistic because it involves an abstract

objective function (who could say what a neuron or a circuit should be trying to optimize?) and a mathematical technique for how to optimize the function (we don't know if neurons can compute gradients of abstract objective functions).

Hence our model is mechanistic, but it does operate at a particular level of abstraction/simplification. We don't model individual ion channels, or biophysics of synaptic plasticity (opening and closing of NMDA channels, accumulation of proteins at synapses, protein synthesis). We do, however, provide a biophysical implementation of the plasticity mechanism through the BDNF/proBDNF model which is more than most models of plasticity achieve, because they typically model a phenomenological STDP or Hebbian rule that just uses activity patterns to potentiate or depress synaptic weights, disregarding how it could be implemented. To the best of our understanding, this is what is normally considered mechanistic in the field (in contrast to, for example, biophysical).

Reviewer #2 (Public Review):

This work combines a model of two-dimensional dendritic growth with attraction and stabilisation by synaptic activity. The authors find that constraining growth models with competition for synaptic inputs produces artificial dendrites that match some key features of real neurons both over development and in terms of final structure. In particular, incorporating distance-dependent competition between synapses of the same dendrite naturally produces distinct phases of dendritic growth (overshoot, pruning, and stabilisation) that are observed biologically and leads to local synaptic organisation with functional relevance. The approach is elegant and well-explained, but makes some significant modelling assumptions that might impact the biological relevance of the results.

Strengths:

The main strength of the work is the general concept of combining morphological models of growth with synaptic plasticity and stabilisation. This is an interesting way to bridge two distinct areas of neuroscience in a manner that leads to findings that could be significant for both. The modelling of both dendritic growth and distance-dependent synaptic competition is carefully done, constrained by reasonable biological mechanisms, and well-described in the text. The paper also links its findings, for example in terms of phases of dendritic growth or final morphological structure, to known data well.

Weaknesses:

The major weaknesses of the paper are the simplifying modelling assumptions that are likely to have an impact on the results. These assumptions are not discussed in enough detail in the current version of the paper.

(1) Axonal dynamics.

A major, and lightly acknowledged, assumption of this paper is that potential synapses, which must come from axons, are fixed in space. This is not realistic for many neural systems, as multiple undifferentiated neurites typically grow from the soma before an axon is specified (Polleux & Snider, 2010). Further, axons are also dynamic structures in early development and, at least in some systems, undergo activity-dependent morphological changes too (O'Leary, 1987; Hall 2000). This paper does not consider the implications of joint pre- and post-synaptic growth and stabilisation.

We thank the reviewer for the summary of the strengths and weaknesses of the work. While we feel that including a full model of axonal dynamics is beyond the scope of the current manuscript, some aspects of axonal dynamics can be included and are now implemented and

tested in the revised manuscript. Since this feedback covers similar aspects of the model that were also pointed out by reviewer #1, we refer here to our detailed reply to their comments 1.1 and 1.2, where we list and discuss all the analyses performed to address the raised issues.

(2) Activity correlations

On a related note, the synapses in the manuscript display correlated activity, but there is no relationship between the distance between synapses and their correlation. In reality, nearby synapses are far more likely to share the same axon and so display correlated activity. If the input activity is spatially correlated and synaptic plasticity displays distance-dependent competition in the dendrites, there is likely to be a non-trivial interaction between these two features with a major impact on the organisation of synaptic contacts onto each neuron.

We have explored the amount of correlation (between and within correlated groups) in the revised manuscript (see also our reply to reviewer comment 1.1).

However, previous experimental work, (e.g. Kleindienst et al., 2011) has provided anatomical and functional analyses that it is unlikely that the functional synaptic clustering on dendritic branches is the result of individual axons making more than one synapse (see pg. 1019).

(3) BDNF dynamics

The models are quite sensitive to the ratio of BDNF to proBDNF (eg Figure 5c). This ratio is also activity-dependent as synaptic activation converts proBDNF into BDNF. The models assume a fixed ratio that is not affected by synaptic activity. There should at least be more justification for this assumption, as there is likely to be a positive feedback relationship between levels of BDNF and synaptic activation.

The reviewer is correct. We used the BDNF-proBDNF model for synaptic plasticity based on our previous work (Kirchner and Gjorgjieva, 2021).

There, we explored only the emergence of functionally clustered synapses on static dendrites which do not grow. In the Methods section (Parameters and data fitting) we justify the choice of the ratio of BDNF to proBDNF from published experimental work. We also performed sensitivity analysis (Supplementary Fig. 1) and perturbation simulations (Supplementary Fig. 3), which showed that the ratio is crucial in regulating the overall amount of potentiation and depression of synaptic efficacy, and therefore has a strong impact on the emergence and maintenance of synaptic organization. Since we already performed all this analysis, we expect that the same results will also apply to the current model which includes dendritic growth, as it involves the same activity-dependent mechanism.

A further weakness is in the discussion of how the final morphologies conform to principles of optimal wiring, which is quite imprecise. 'Optimal wiring' in the sense of dendrites and axons (Cajal, 1895; Chklovskii, 2004; Cuntz et al, 2007, Budd et al, 2010) is not usually synonymous with 'shortest wiring' as implied here. Instead, there is assumed to be a balance between minimising total dendritic length and minimising the tree distance (ie Figure 4c here) between synapses and the site of input integration, typically the soma. The level of this balance gives the deviation from the theoretical minimum length as direct paths to synapses typically require longer dendrites. In the model this is generated by the guidance of dendritic growth directly towards the synaptic targets. The interpretation of the deviation in this results section discussing optimal wiring, with hampered diffusion of signalling molecules, does not seem to be correct.

We agree with this comment. We had wrongly used the term “optimal wiring” as neurons can optimize their wiring not only by minimizing their dendritic length but other factors as noted by the reviewer. In the revised manuscript we replaced the term “optimal wiring” with “minimal wiring” wherever it was incorrectly used. On top of that, we performed further analysis and discussed these differences, as pointed out in the reply to reviewer #1 point 1.8.

To summarize, we want to again thank the reviewer for their in-depth review and all the suggestions that helped us improve the analysis and implementation of our model.

Reviewer #3 (Public Review):

The authors propose a mechanistic model of how the interplay between activity-independent growth and an activity-dependent synaptic strengthening/weaken model influences the dendrite shape, complexity and distribution of synapses. The authors focus on a model for stellate cells, which have multiple dendrites emerging from a soma. The activity independent component is provided by a random pool of presynaptic sites that represent potential synapses and that release a diffusible signal that promotes dendritic growth. Then a spontaneous activity pattern with some correlation structure is imposed at those presynaptic sites. The strength of these synapses follow a learning rule previously proposed by the lab: synapses strengthen when there is correlated firing across multiple sites, and synapses weaken if there is uncorrelated firing with the relative strength of these processes controlled by available levels of BDNF/proBDNF. Once a synapse is weakened below a threshold, the dendrite branch at that site retracts and loses its sensitivity to the growth signal

The authors run the simulation and map out how dendrites and synapses evolve and stabilize. They show that dendritic trees growing rapidly and then stabilize by balancing growth and retraction (Figure 2). They also that there is an initial bout of synaptogenesis followed by loss of synapses, reflecting the longer amount of time it takes to weaken a synapse (Figure 3). They analyze how this evolution of dendrites and synapses depends on the correlated firing of synapses (i.e. defined as being in the same "activity group"). They show that in the stabilized phase, synapses that remain connected to a given dendritic branch are likely to be from same activity group (Figure 4). The authors systemically alter the learning rule by changing the available concentration of BDNF, which alters the relative amount of synaptic strengthening, which in turn affects stabilization, density of synapses and interestingly how selective for an activity group one dendrite is (Figure 5). In addition the authors look at how altering the activity-independent factors influences outgrowth (Figure 6). Finally, one of the interesting outcomes is that the resulting dendritic trees represent "optimal wiring" solutions in the sense that dendrites use the shortest distance given the distribution of synapses. They compare this distribute to one published data to see how the model compared to what has been observed experimentally.

There are many strengths to this study. The consequence of adding the activity-dependent contribution to models of synapto- and dendritogenesis is novel. There is some exploration of parameters space with the motivation of keeping the parameters as well as the generated outcomes close to anatomical data of real dendrites. The paper is also scholarly in its comparison of this approach to previous generative models. This work represented an important advance to our understanding of how learning rules can contribute to dendrite morphogenesis.

We thank the reviewer for the positive evaluation of the work and the suggestions below.

To improve the clarity of the manuscript, we adjusted and fixed some figures and corresponding paragraphs as follows:

- (1) We increased the number of ticks and their corresponding numbers in all the figures to make them easier to read and interpret.
- (2) In Figure 3 panel d, showing the evolution of synaptic weight, we corrected the upper limit at the y-axis to 1 (from previously 2).
- (3) Due to a typo in the implementation of the BDNF concentration, we had to correct the used BDNF concentrations from 49%, 45% and 40%, to 49%, 46.5% and 43% respectively.
- (4) The y-axis labels of Figure 6 (old Figure 5) panel e and f were changed to make the plots clearer (e: “morphology change explained (%)” to “effect on morphology (%)”, and f: “synapse connection explained (%)” to “effect on connected synapses (%)”).
- (5) The values for the eta and tau-w in the supplementary Table were corrected. Previously tau-w was falsely 6000 time steps which was corrected to 3000 time steps, and eta was 45% and is now 46.5%.

We believe that all the changes to the manuscript will address the reviewer’s concerns and enhance the clarity and accuracy of the findings described in the manuscript.

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